

last time

locks

deadlock

avoidance by

- don't hold resource while acquiring another

- abort and retry

- consistent order

barriers

compute minimum of 100M element array with 2 processors

algorithm:

compute minimum of 50M of the elements on each CPU

one thread for each CPU

wait for all computations to finish

take minimum of all the minimums

barriers

compute minimum of 100M element array with 2 processors

algorithm:

compute minimum of 50M of the elements on each CPU
one thread for each CPU

wait for all computations to finish

take minimum of all the minimums

barriers API

`barrier.Initialize(NumberOfThreads)`

`barrier.Wait()` — return after all threads have waited

idea: multiple threads perform computations in parallel

threads wait for **all other threads** to call `Wait()`

barrier: waiting for finish

```
barrier.Initialize(2);
```

Thread 0

```
partial_mins[0] =  
    /* min of first  
       50M elems */;
```

```
barrier.Wait();
```

```
total_min = min(  
    partial_mins[0],  
    partial_mins[1]  
);
```

Thread 1

```
partial_mins[1] =  
    /* min of last  
       50M elems */  
barrier.Wait();
```

barriers: reuse

Thread 0

```
results[0][0] = getInitial(0);  
barrier.Wait();
```

```
results[1][0] =  
    computeFrom(0,  
        results[0][0],  
        results[0][1]  
    );  
barrier.Wait();
```

```
results[2][0] =  
    computeFrom(0,  
        results[1][0],  
        results[1][1]  
    );
```

Thread 1

```
results[0][1] = getInitial(1);  
barrier.Wait();
```

```
results[1][1] =  
    computeFrom(1,  
        results[0][0],  
        results[0][1]  
    );  
barrier.Wait();
```

```
results[2][1] =  
    computeFrom(1,  
        results[1][0],  
        results[1][1]  
    );
```

barriers: reuse

Thread 0

```
results[0][0] = getInitial(0);  
barrier.Wait();
```

```
results[1][0] =  
    computeFrom(0,  
        results[0][0],  
        results[0][1]  
    );  
barrier.Wait();
```

```
results[2][0] =  
    computeFrom(0,  
        results[1][0],  
        results[1][1]  
    );
```

Thread 1

```
results[0][1] = getInitial(1);  
barrier.Wait();
```

```
results[1][1] =  
    computeFrom(1,  
        results[0][0],  
        results[0][1]  
    );  
barrier.Wait();
```

```
results[2][1] =  
    computeFrom(1,  
        results[1][0],  
        results[1][1]  
    );
```


barriers: reuse

Thread 0

```
results[0][0] = getInitial(0);  
barrier.Wait();
```

```
results[1][0] =  
    computeFrom(0,  
        results[0][0],  
        results[0][1]  
    );  
barrier.Wait();
```

```
results[2][0] =  
    computeFrom(0,  
        results[1][0],  
        results[1][1]  
    );
```

Thread 1

```
results[0][1] = getInitial(1);  
barrier.Wait();
```

```
results[1][1] =  
    computeFrom(1,  
        results[0][0],  
        results[0][1]  
    );  
barrier.Wait();
```

```
results[2][1] =  
    computeFrom(1,  
        results[1][0],  
        results[1][1]  
    );
```

pthread barriers

```
pthread_barrier_t barrier;  
pthread_barrier_init(  
    &barrier,  
    NULL /* attributes */,  
    numberOfThreads  
);  
...  
...  
pthread_barrier_wait(&barrier);
```

exercise

```
pthread_barrier_t barrier; int x = 0, y = 0;
void thread_one() {
    y = 10;
    pthread_barrier_wait(&barrier);
    y = x + y;
    pthread_barrier_wait(&barrier);
    pthread_barrier_wait(&barrier);
    printf("%d_%d\n", x, y);
}
void thread_two() {
    x = 20;
    pthread_barrier_wait(&barrier);
    pthread_barrier_wait(&barrier);
    x = x + y;
    pthread_barrier_wait(&barrier);
}
```

output? (if both run at once, barrier set for 2 threads)

life homework (pseudocode)

```
for (int time = 0; time < MAX_ITERATIONS; ++time) {  
    for (int y = 0; y < size; ++y) {  
        for (int x = 0; x < size; ++x) {  
            to_grid(x, y) = computeValue(from_grid, x, y);  
        }  
    }  
    swap(from_grid, to_grid);  
}
```

life homework

compute grid of values for time t from grid for time $t - 1$

compute new value at i, j based on surrounding values

parallel version: produce parts of grid in different threads

use barriers to finish time t before going to time $t + 1$

monitors/condition variables

locks for mutual exclusion

condition variables for waiting for event

represents **list of waiting threads**

operations: wait (for event); signal/broadcast (that event happened)

related data structures

monitor = lock + 0 or more condition variables + shared data

Java: every object is a monitor (has instance variables, built-in lock, cond. var)

python: build your own: provides you locks + condition variables

monitor idea

a monitor

lock
shared data
condvar 1
condvar 2
...
operation1(...)
operation2(...)

monitor idea

a monitor

lock
shared data
condvar 1
condvar 2
...
operation1(...)
operation2(...)

lock must be acquired
before accessing
any part of monitor's stuff

monitor idea

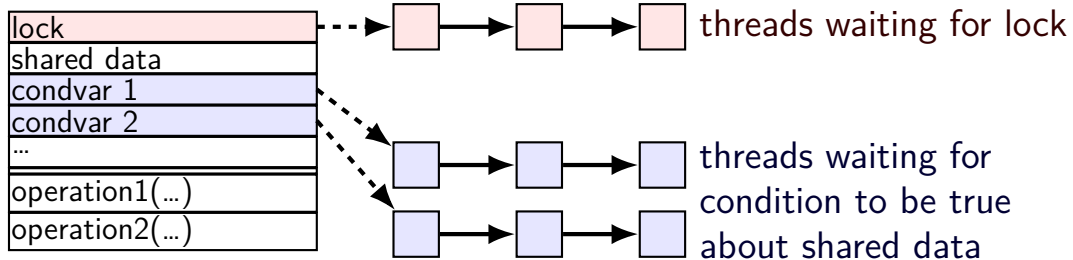
a monitor

lock
shared data
condvar 1
condvar 2
...
operation1(...)
operation2(...)



monitor idea

a monitor



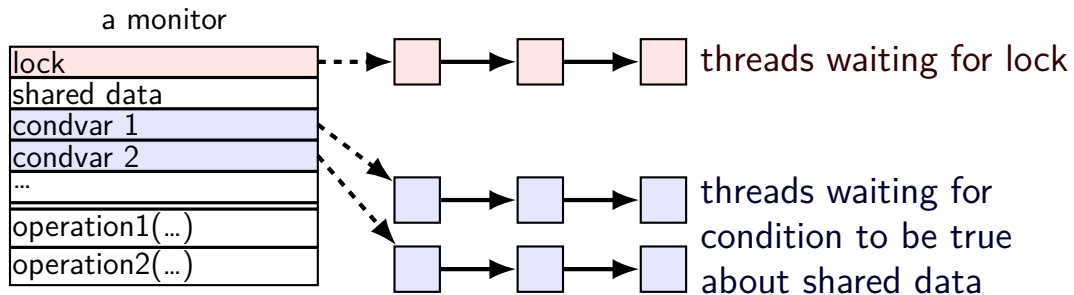
condvar operations

condvar operations:

Wait(cv, lock) — unlock lock, add current thread to cv queue
...and reacquire lock before returning

Broadcast(cv) — remove all from condvar queue

Signal(cv) — remove one from condvar queue



condvar operations

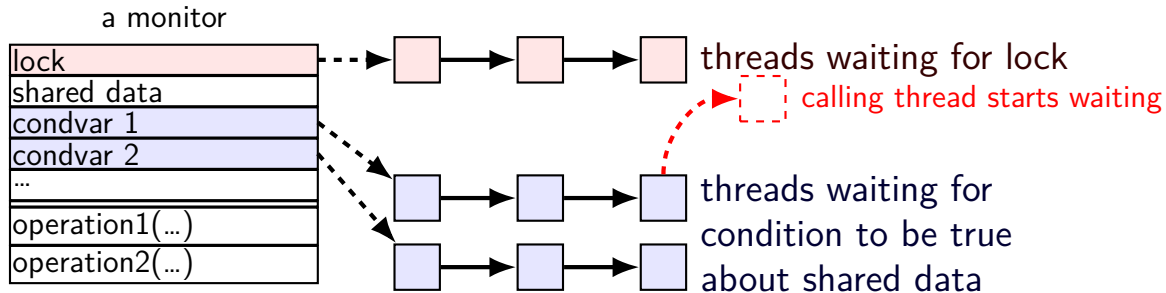
condvar operations:

Wait(cv, lock) — unlock lock, add current thread to cv queue

...and reacquire lock before returning

Broadcast(cv) — remove all from condvar queue

Signal(cv) — remove one from condvar queue



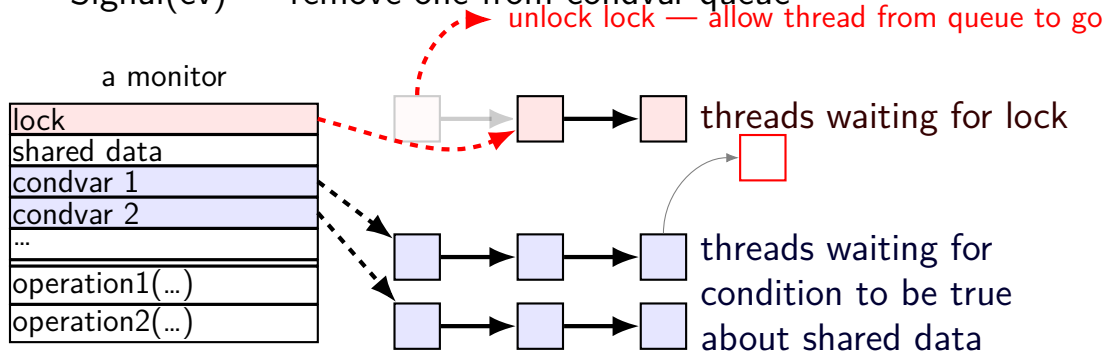
condvar operations

condvar operations:

Wait(cv, lock) — **unlock** lock, add current thread to cv queue
...and **reacquire** lock before returning

Broadcast(cv) — remove all from condvar queue

Signal(cv) — remove one from condvar queue



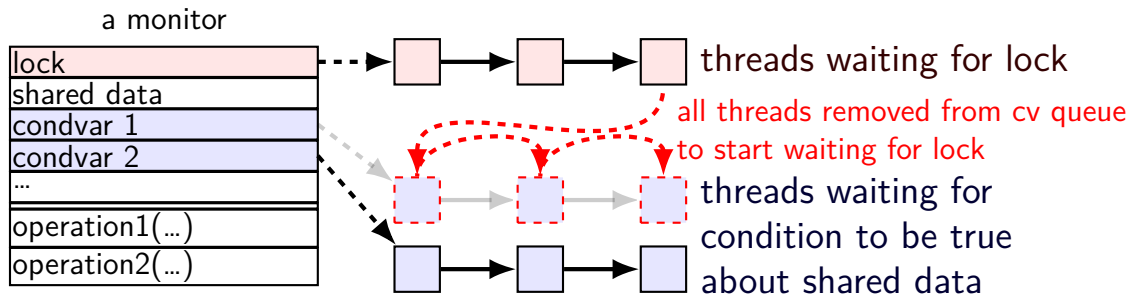
condvar operations

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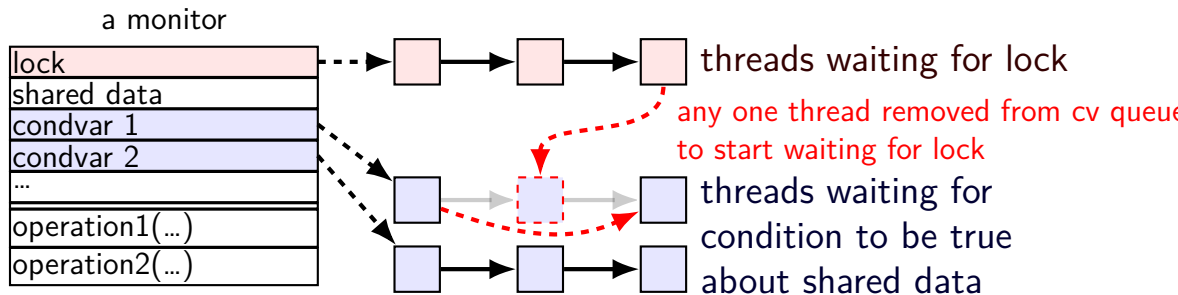
condvar operations

condvar operations:

Wait(cv, lock) — unlock lock, add current thread to cv queue
...and reacquire lock before returning

Broadcast(cv) — remove all from condvar queue

Signal(cv) — remove one from condvar queue



pthread cv usage

// MISSING: init calls, etc.

```
pthread_mutex_t lock;  
bool finished;    // data, only accessed with after acquiring lock  
pthread_cond_t finished_cv; // to wait for 'finished' to be true
```

```
void WaitForFinished() {  
    pthread_mutex_lock(&lock);  
    while (!finished) {  
        pthread_cond_wait(&finished_cv, &lock);  
    }  
    pthread_mutex_unlock(&lock);  
}
```

```
void Finish() {  
    pthread_mutex_lock(&lock);  
    finished = true;  
    pthread_cond_broadcast(&finished_cv);  
    pthread_mutex_unlock(&lock);  
}
```


pthread cv usage

// MISSING: init calls, etc.

pthread_mutex_t lock;

bool finished; *// data, only accessed with after acquiring lock*

pthread_cond_t finished_cv; *// to wait for 'finished' to be true*

void WaitForFinished() {

pthread_mutex_lock(&lock);

while (!finished) {

pthread_cond_wait(&finished_cv, &lock);

}

pthread_mutex_unlock(&lock);

}

acquire lock before
reading or writing finished

void Finish() {

pthread_mutex_lock(&lock);

finished = true;

pthread_cond_broadcast(&finished_cv);

pthread_mutex_unlock(&lock);

}

pthread cv usage

// MISSING: init calls, etc.

```
pthread_mutex_t lock;
```

```
bool finished;    // data, only accessed with after acquiring lock
```

```
pthread_cond_t finished_cv; // to wait for 'finished' to be true
```

```
void WaitForFinished() {  
    pthread_mutex_lock(&lock);  
    while (!finished) {  
        pthread_cond_wait(&finished_cv, &lock);  
    }  
    pthread_mutex_unlock(&lock);  
}
```

check whether we need to wait at all
(why a loop? we'll explain later)

```
void Finish() {  
    pthread_mutex_lock(&lock);  
    finished = true;  
    pthread_cond_broadcast(&finished_cv);  
    pthread_mutex_unlock(&lock);  
}
```

pthread cv usage

// MISSING: init calls, etc.

```
pthread_mutex_t lock;
```

```
bool finished;    // data, only accessed with after acquiring lock
```

```
pthread_cond_t finished_cv; // to wait for 'finished' to be true
```

```
void WaitForFinished() {
```

```
    pthread_mutex_lock(&lock);
```

```
    while (!finished) {
```

```
        pthread_cond_wait(&finished_cv, &lock);
```

```
    }
```

```
    pthread_mutex_unlock(&lock);
```

```
}
```

```
void Finish() {
```

```
    pthread_mutex_lock(&lock);
```

```
    finished = true;
```

```
    pthread_cond_broadcast(&finished_cv);
```

```
    pthread_mutex_unlock(&lock);
```

```
}
```

know we need to wait

(finished can't change while we have lock)

so wait, releasing lock...

pthread cv usage

// MISSING: init calls, etc.

```
pthread_mutex_t lock;
```

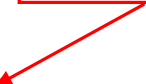
```
bool finished;    // data, only accessed with after acquiring lock
```

```
pthread_cond_t finished_cv; // to wait for 'finished' to be true
```

```
void WaitForFinished() {  
    pthread_mutex_lock(&lock);  
    while (!finished) {  
        pthread_cond_wait(&finished_cv, &lock);  
    }  
    pthread_mutex_unlock(&lock);  
}
```

```
void Finish() {  
    pthread_mutex_lock(&lock);  
    finished = true;  
    pthread_cond_broadcast(&finished_cv);  
    pthread_mutex_unlock(&lock);  
}
```

allow all waiters to proceed
(once we unlock the lock)



WaitForFinish timeline 1

WaitForFinish thread	Finish thread
<code>mutex_lock(&lock)</code> (thread has lock)	
	<code>mutex_lock(&lock)</code> (start waiting for lock)
<code>while (!finished) ...</code> <code>cond_wait(&finished_cv, &lock);</code> (start waiting for cv)	(done waiting for lock)
	<code>finished = true</code> <code>cond_broadcast(&finished_cv)</code>
(done waiting for cv) (start waiting for lock)	
	<code>mutex_unlock(&lock)</code>
(done waiting for lock) <code>while (!finished) ...</code> (finished now true, so return) <code>mutex_unlock(&lock)</code>	

WaitForFinish timeline 2

WaitForFinish thread	Finish thread
	<code>mutex_lock(&lock)</code> <code>finished = true</code> <code>cond_broadcast(&finished_cv)</code> <code>mutex_unlock(&lock)</code>
<code>mutex_lock(&lock)</code> <code>while (!finished) ...</code> (finished now true, so return) <code>mutex_unlock(&lock)</code>	

why the loop

```
while (!finished) {  
    pthread_cond_wait(&finished_cv, &lock);  
}
```

we only broadcast if finished is true

so why check finished afterwards?

why the loop

```
while (!finished) {  
    pthread_cond_wait(&finished_cv, &lock);  
}
```

we only broadcast if finished is true

so why check finished afterwards?

pthread_cond_wait manual page:

“**Spurious wakeups** ... may occur.”

spurious wakeup = wait returns even though nothing happened

unbounded buffer producer/consumer

```
pthread_mutex_t lock;  
pthread_cond_t data_ready;  
UnboundedQueue buffer;
```

```
Produce(item) {  
    pthread_mutex_lock(&lock);  
    buffer.enqueue(item);  
    pthread_cond_signal(&data_ready);  
    pthread_mutex_unlock(&lock);  
}  
Consume() {  
    pthread_mutex_lock(&lock);  
    while (buffer.empty()) {  
        pthread_cond_wait(&data_ready, &lock);  
    }  
    item = buffer.dequeue();  
    pthread_mutex_unlock(&lock);  
    return item;  
}
```

unbounded buffer producer/consumer

```
pthread_mutex_t lock;  
pthread_cond_t data_ready;  
UnboundedQueue buffer;
```

```
Produce(item) {  
    pthread_mutex_lock(&lock);  
    buffer.enqueue(item);  
    pthread_cond_signal(&data_ready);  
    pthread_mutex_unlock(&lock);  
}
```

```
Consume() {  
    pthread_mutex_lock(&lock);  
    while (buffer.empty()) {  
        pthread_cond_wait(&data_ready, &lock);  
    }  
    item = buffer.dequeue();  
    pthread_mutex_unlock(&lock);  
    return item;  
}
```

rule: never touch buffer
without acquiring lock

otherwise: what if two threads
simultaneously en/dequeue?
(both use same array/linked list entry?)
(both reallocate array?)

unbounded buffer producer/consumer

```
pthread_mutex_t lock;  
pthread_cond_t data_ready;  
UnboundedQueue buffer;
```

```
Produce(item) {  
    pthread_mutex_lock(&lock);  
    buffer.enqueue(item);  
    pthread_cond_signal(&data_ready);  
    pthread_mutex_unlock(&lock);  
}
```

```
Consume() {  
    pthread_mutex_lock(&lock);  
    while (buffer.empty()) {  
        pthread_cond_wait(&data_ready, &lock);  
    }  
    item = buffer.dequeue();  
    pthread_mutex_unlock(&lock);  
    return item;  
}
```

check if empty
if so, dequeue

okay because have lock

other threads cannot dequeue here

unbounded buffer producer/consumer

```
pthread_mutex_t lock;  
pthread_cond_t data_ready;  
UnboundedQueue buffer;
```

```
Produce(item) {  
    pthread_mutex_lock(&lock);  
    buffer.enqueue(item);  
    pthread_cond_signal(&data_ready);  
    pthread_mutex_unlock(&lock);  
}  
Consume() {  
    pthread_mutex_lock(&lock);  
    while (buffer.empty()) {  
        pthread_cond_wait(&data_ready, &lock);  
    }  
    item = buffer.dequeue();  
    pthread_mutex_unlock(&lock);  
    return item;  
}
```

wake one Consume thread
if any are waiting



unbounded buffer producer/consumer

```
pthread_mutex_t lock;  
pthread_cond_t data_ready;  
UnboundedQueue buffer;
```

```
Produce(item) {  
    pthread_mutex_lock(&lock);  
    buffer.enqueue(item);  
    pthread_cond_signal(&data_ready);  
    pthread_mutex_unlock(&lock);  
}
```

```
Consume() {  
    pthread_mutex_lock(&lock);  
    while (buffer.empty()) {  
        pthread_cond_wait(&data_ready, &lock);  
    }  
    item = buffer.dequeue();  
    pthread_mutex_unlock(&lock);  
    return item;  
}
```

Thread 1

Produce()
...lock
...enqueue
...signal
...unlock

Thread 2

Consume()
...lock
...empty? no
...dequeue
...unlock
return

0 iterations: Produce() called before Consume()
1 iteration: Produce() signalled, probably
2+ iterations: spurious wakeup or ...?

unbounded buffer producer/consumer

```
pthread_mutex_t lock;  
pthread_cond_t data_ready;  
UnboundedQueue buffer;
```

```
Produce(item) {  
    pthread_mutex_lock(&lock);  
    buffer.enqueue(item);  
    pthread_cond_signal(&data_ready);  
    pthread_mutex_unlock(&lock);  
}
```

```
Consume() {  
    pthread_mutex_lock(&lock);  
    while (buffer.empty()) {  
        pthread_cond_wait(&data_ready, &lock);  
    }  
    item = buffer.dequeue();  
    pthread_mutex_unlock(&lock);  
    return item;  
}
```

Thread 1

Thread 2

	Consume()
	...lock
	...empty? yes
	...unlock/start wait
Produce()	waiting for data_ready
...lock	
...enqueue	
...signal	stop wait
...unlock	lock
	...empty? no
	...dequeue
	...unlock
	return

0 iterations: Produce() called before Consume()
1 iteration: Produce() signalled, probably
2+ iterations: spurious wakeup or ...?

unbounded buffer producer/consumer

```
pthread_mutex_t lock;
pthread_cond_t data_ready;
UnboundedQueue buffer;
```

```
Produce(item) {
    pthread_mutex_lock(&lock);
    buffer.enqueue(item);
    pthread_cond_signal(&data_ready);
    pthread_mutex_unlock(&lock);
}
```

```
Consume() {
    pthread_mutex_lock(&lock);
    while (buffer.empty()) {
        pthread_cond_wait(&data_ready, &lock);
    }
    item = buffer.dequeue();
    pthread_mutex_unlock(&lock);
    return item;
}
```

Thread 1

Produce()
...lock
...enqueue
...signal
...unlock

Thread 2

Consume()
...lock
...empty? yes
...unlock/start wait
waiting for data_ready
stop wait
waiting for lock
...lock
...empty? yes
...unlock/start wait

Thread 3

Consume()
waiting for lock
lock
...empty? no
...dequeue
...unlock
return

0 iterations: Produce() called before Consume()
 1 iteration: Produce() signalled, probably
 2+ iterations: spurious wakeup or ...?

unbounded buffer producer/consumer

```
pthread_mutex_t lock;
pthread_cond_t data_ready;
UnboundedQueue buffer;
```

in pthreads: signalled thread not
guaranteed to hold lock next

alternate design:
signalled thread gets lock next
called "Hoare scheduling"
not done by pthreads, Java, ...

```
pthread_cond_wait(&data_ready, &lock);
}
item = buffer.dequeue();
pthread_mutex_unlock(&lock);
return item;
}
```

Thread 1

```
Produce()
...lock
...enqueue
...signal
...unlock
```

Thread 2

```
Consume()
...lock
...empty? yes
...unlock/start wait

waiting for
data_ready

stop wait

waiting for
lock

...lock
...empty? yes
...unlock/start wait
```

Thread 3

```
Consume()
waiting for
lock

lock
...empty? no
...dequeue
...unlock
return
```

0 iterations: Produce() called before Consume()
1 iteration: Produce() signalled, probably
2+ iterations: spurious wakeup or ...?

Hoare versus Mesa monitors

Hoare-style monitors

- signal 'hands off' lock to awoken thread

Mesa-style monitors

- any eligible thread gets lock next
(maybe some other idea of priority?)

every current threading library I know of does Mesa-style

bounded buffer producer/consumer

```
pthread_mutex_t lock;
pthread_cond_t data_ready; pthread_cond_t space_ready;
BoundedQueue buffer;
Produce(item) {
    pthread_mutex_lock(&lock);
    while (buffer.full()) { pthread_cond_wait(&space_ready, &lock); }
    buffer.enqueue(item);
    pthread_cond_signal(&data_ready);
    pthread_mutex_unlock(&lock);
}
Consume() {
    pthread_mutex_lock(&lock);
    while (buffer.empty()) {
        pthread_cond_wait(&data_ready, &lock);
    }
    item = buffer.dequeue();
    pthread_cond_signal(&space_ready);
    pthread_mutex_unlock(&lock);
    return item;
}
```

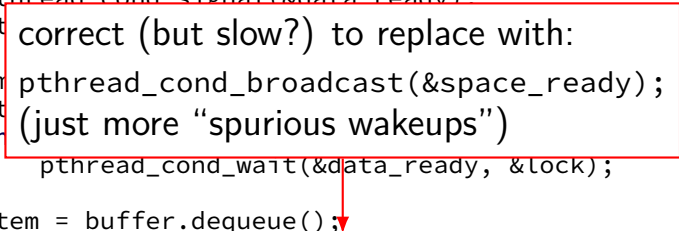
bounded buffer producer/consumer

```
pthread_mutex_t lock;
pthread_cond_t data_ready; pthread_cond_t space_ready;
BoundedQueue buffer;
Produce(item) {
    pthread_mutex_lock(&lock);
    while (buffer.full()) { pthread_cond_wait(&space_ready, &lock); }
    buffer.enqueue(item);
    pthread_cond_signal(&data_ready);
    pthread_mutex_unlock(&lock);
}
Consume() {
    pthread_mutex_lock(&lock);
    while (buffer.empty()) {
        pthread_cond_wait(&data_ready, &lock);
    }
    item = buffer.dequeue();
    pthread_cond_signal(&space_ready);
    pthread_mutex_unlock(&lock);
    return item;
}
```

bounded buffer producer/consumer

```
pthread_mutex_t lock;
pthread_cond_t data_ready; pthread_cond_t space_ready;
BoundedQueue buffer;
Produce(item) {
    pthread_mutex_lock(&lock);
    while (buffer.full()) { pthread_cond_wait(&space_ready, &lock); }
    buffer.enqueue(item);
    pthread_cond_signal(&data_ready);
}
Consumption {
    pthread_cond_wait(&data_ready, &lock);
    item = buffer.dequeue();
    pthread_cond_signal(&space_ready);
    pthread_mutex_unlock(&lock);
    return item;
}
```

correct (but slow?) to replace with:
pthread_cond_broadcast(&space_ready);
(just more “spurious wakeups”)



bounded buffer producer/consumer

```
pthread_mutex_t lock;
pthread_cond_t data_ready; pthread_cond_t space_ready;
BoundedQueue buffer;
Produce(item) {
    pthread_mutex_lock(&lock);
    while (buffer.full()) { pthread_cond_wait(&space_ready, &lock); }
    buffer.enqueue(item);
    pthread_cond_signal(&data_ready);
    pthread_mutex_unlock(&lock);
}
Consume() {
    pthread_mutex_lock(&lock);
    while (buffer.empty()) {
        pthread_cond_wait(&data_ready, &lock);
    }
    item = buffer.dequeue();
    pthread_cond_signal(&space_ready);
    pthread_mutex_unlock(&lock);
    return item;
}
```

correct but slow to replace
data_ready and space_ready
with 'combined' condvar ready
and use broadcast
(just more "spurious wakeups")

monitor pattern

```
pthread_mutex_lock(&lock);
while (!condition A) {
    pthread_cond_wait(&condvar_for_A, &lock);
}
... /* manipulate shared data, changing other conditions */
if (set condition A) {
    pthread_cond_broadcast(&condvar_for_A);
    /* or signal, if only one thread cares */
}
if (set condition B) {
    pthread_cond_broadcast(&condvar_for_B);
    /* or signal, if only one thread cares */
}
...
pthread_mutex_unlock(&lock)
```

monitors rules of thumb

never touch shared data without holding the lock

keep lock held for **entire operation**:

verifying condition (e.g. buffer not full) *up to and including*
manipulating data (e.g. adding to buffer)

create condvar for every kind of scenario waited for

always write **loop** calling cond_wait to wait for condition X

broadcast/signal condition variable **every time you change X**

monitors rules of thumb

never touch shared data without holding the lock

keep lock held for **entire operation**:

verifying condition (e.g. buffer not full) *up to and including*
manipulating data (e.g. adding to buffer)

create condvar for every kind of scenario waited for

always write **loop** calling cond_wait to wait for condition X

broadcast/signal condition variable **every time you change X**

correct but slow to...

broadcast when just signal would work

broadcast or signal when nothing changed

use one condvar for multiple conditions

mutex/cond var init/destroy

```
pthread_mutex_t mutex;  
pthread_cond_t cv;  
pthread_mutex_init(&mutex, NULL);  
pthread_cond_init(&cv, NULL);  
// --OR--  
pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;  
pthread_cond_t cv = PTHREAD_COND_INITIALIZER;  
  
// and when done:  
...  
pthread_cond_destroy(&cv);  
pthread_mutex_destroy(&mutex);
```

wait for both finished

```
// MISSING: init calls, etc.
```

```
pthread_mutex_t lock;  
bool finished[2];  
pthread_cond_t both_finished_cv;
```

```
void WaitForBothFinished() {  
    pthread_mutex_lock(&lock);  
    while (_____) {  
        pthread_cond_wait(&both_finished_cv, &lock);  
    }  
    pthread_mutex_unlock(&lock);  
}
```

```
void Finish(int index) {  
    pthread_mutex_lock(&lock);  
    finished[index] = true;  
    -----  
    pthread_mutex_unlock(&lock);  
}
```

wait for both finished

// MISSING: init calls, etc.

```
pthread_mutex_t lock;  
bool finished[2];  
pthread_cond_t both_finished_cv;
```

```
void WaitForBothFinished() {  
    pthread_mutex_lock(&lock);  
    while ( ) {  
        pthread_cond_wait(&both_finished_cv, &lock);  
    }  
    pthread_mutex_unlock(&lock);  
}
```

```
void Finish(int index) {  
    pthread_mutex_lock(&lock);  
    finished[index] = true;  
    -----  
    pthread_mutex_unlock(&lock);  
}
```

- A. `finished[0] && finished[1]`
- B. `finished[0] || finished[1]`
- C. `!finished[0] || !finished[1]`
- D. `finished[0] != finished[1]`
- E. something else

wait for both finished

// MISSING: init calls, etc.

```
pthread_mutex_t lock;  
bool finished[2];  
pthread_cond_t both_finished;
```

```
void WaitForBothFinished
```

```
pthread_mutex_lock(&lock)
```

```
while (_____
```

```
pthread_cond_wait(&both_finished_cv, &lock);
```

```
}  
pthread_mutex_unlock(&lock);
```

```
}
```

```
void Finish(int index) {
```

```
pthread_mutex_lock(&lock);
```

```
finished[index] = true;
```

```
pthread_mutex_unlock(&lock);
```

```
}
```

- A. pthread_cond_signal(&both_finished_cv)
- B. pthread_cond_broadcast(&both_finished_cv)
- C. if (finished[1-index])
 pthread_cond_signal(&both_finished_cv);
- D. if (finished[1-index])
 pthread_cond_broadcast(&both_finished_cv);
- E. something else

monitor exercise: one-use barrier

suppose we want to implement a one-use barrier; fill in blanks:

```
struct BarrierInfo {
    pthread_mutex_t lock;
    int total_threads; // initially total # of threads
    int number_reached; // initially 0
    -----
};
void BarrierWait(BarrierInfo *b) {
    pthread_mutex_lock(&b->lock);
    ++b->number_reached;
    if (b->number_reached == b->total_threads) {
        -----
    } else {
        -----
        -----
    }
    pthread_mutex_unlock(&b->lock);
}
```

generalizing locks: semaphores

semaphore has a non-negative integer **value** and two operations:

P() or **down** or **wait**:

wait for semaphore to become positive (> 0),
then decrement by 1

V() or **up** or **signal** or **post**:

increment semaphore by 1 (waking up thread if needed)

P, V from Dutch: *proberen* (test), *verhogen* (increment)

semaphores are kinda integers

semaphore like an integer, but...

cannot read/write directly

down/up operation only way to access (typically)

exception: initialization

never negative — wait instead

down operation wants to make negative? thread waits

reserving books

suppose tracking copies of library book...

```
Semaphore free_copies = Semaphore(3);  
void ReserveBook() {  
    // wait for copy to be free  
    free_copies.down();  
    ... // ... then take reserved copy  
}  
  
void ReturnBook() {  
    ... // return reserved copy  
    free_copies.up();  
    // ... then wakeup waiting thread  
}
```


counting resources: reserving books

suppose tracking copies of same library book

non-negative integer count = # how many books used?

up = give back book; down = take book

Copy 1
Copy 2
Copy 3

free copies

3

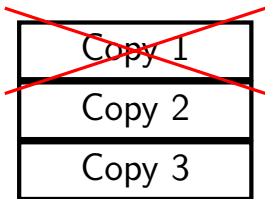
counting resources: reserving books

suppose tracking copies of same library book

non-negative integer count = # how many books used?

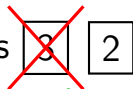
up = give back book; down = take book

taken out



Copy 1
Copy 2
Copy 3

free copies



after calling down to reserve

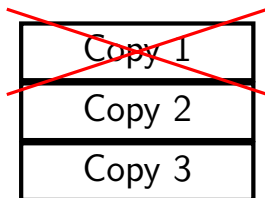
counting resources: reserving books

suppose tracking copies of same library book

non-negative integer count = # how many books used?

up = give back book; down = take book

taken out



Copy 1
Copy 2
Copy 3

free copies 2

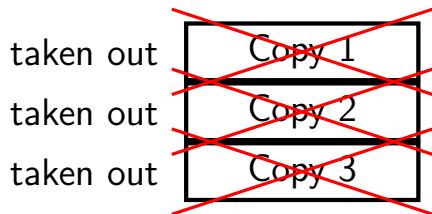
after calling down to reserve

counting resources: reserving books

suppose tracking copies of same library book

non-negative integer count = # how many books used?

up = give back book; down = take book



free copies 0

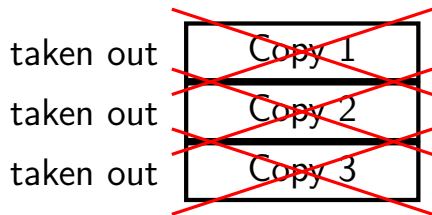
after calling down three times
to reserve all copies

counting resources: reserving books

suppose tracking copies of same library book

non-negative integer count = # how many books used?

up = give back book; down = take book



free copies

0



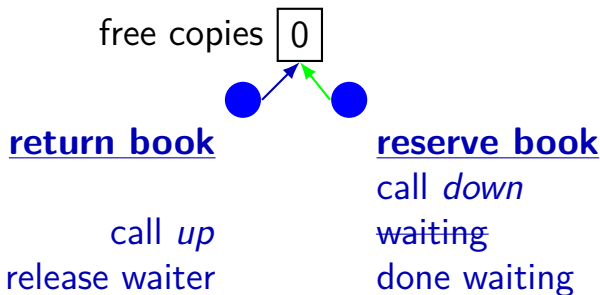
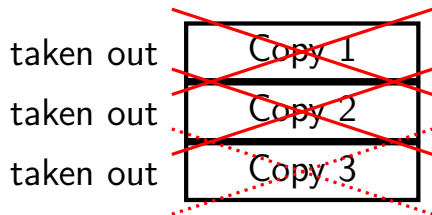
reserve book
call *down* again
start waiting...

counting resources: reserving books

suppose tracking copies of same library book

non-negative integer count = # how many books used?

up = give back book; **down** = take book



modifying cache blocks in parallel

typical memory access — less than cache block

e.g. one 4-byte array element in 64-byte cache block

what if two processors modify different parts same cache block?

4-byte writes to 64-byte cache block

typically how caches work — write instructions happen one at a time:

processor 'locks' 64-byte cache block, fetching latest version

processor updates 4 bytes of 64-byte cache block

later, processor might give up cache block

modifying things in parallel (code)

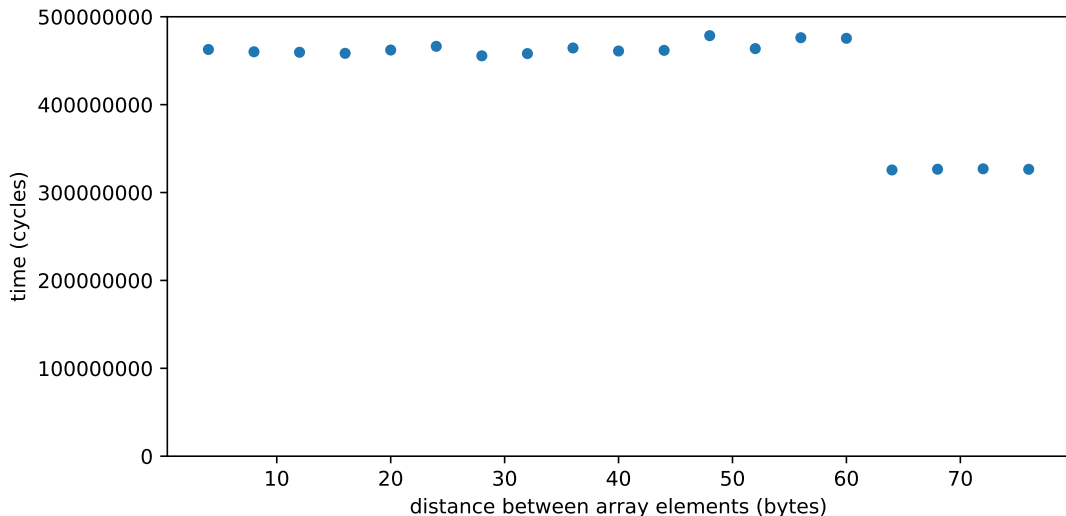
```
void *sum_up(void *raw_dest) {
    int *dest = (int *) raw_dest;
    for (int i = 0; i < 64 * 1024 * 1024; ++i) {
        *dest += data[i];
    }
}

__attribute__((aligned(4096)))
int dests[1024]; /* aligned = address is mult. of 4096 */

void sum_twice(int distance) {
    pthread_t threads[2];
    pthread_create(&threads[0], NULL, sum_up, &dests[0]);
    pthread_create(&threads[1], NULL, sum_up, &dests[distance]);
    pthread_join(threads[0], NULL);
    pthread_join(threads[1], NULL);
}
```


performance v. array element gap

(assuming `sum_up` compiled to not omit memory accesses)



false sharing

synchronizing to access two independent things

two parts of same cache block

solution: separate them

exercise (1)

```
int values[1024]; int results[2];
void *sum_front(void *ignored_argument) {
    results[0] = 0;
    for (int i = 0; i < 512; ++i)
        results[0] += values[i];
    return NULL;
}
void *sum_back(void *ignored_argument) {
    results[1] = 0;
    for (int i = 512; i < 1024; ++i)
        results[1] += values[i];
    return NULL;
}
int sum_all() {
    pthread_t sum_front_thread, sum_back_thread;
    pthread_create(&sum_front_thread, NULL, sum_front, NULL);
    pthread_create(&sum_back_thread, NULL, sum_back, NULL);
    pthread_join(sum_front_thread, NULL);
    pthread_join(sum_back_thread, NULL);
    return results[0] + results[1];
}
```

Where is false sharing likely to occur? How to fix?

exercise (2)

```
struct ThreadInfo { int *values; int start; int end; int result };
void *sum_thread(void *argument) {
    ThreadInfo *my_info = (ThreadInfo *) argument;
    int sum = 0;
    for (int i = my_info->start; i < my_info->end; ++i) {
        my_info->result += my_info->values[i];
    }
    return NULL;
}

int sum_all(int *values) {
    ThreadInfo info[2]; pthread_t thread[2];
    for (int i = 0; i < 2; ++i) {
        info[i].values = values; info[i].start = i*512; info[i].end = (i+1)*512;
        pthread_create(&threads[i], NULL, sum_thread, (void *) &info[i]);
    }
    for (int i = 0; i < 2; ++i)
        pthread_join(threads[i], NULL);
    return info[0].result + info[1].result;
}
```

Where is false sharing likely to occur?

transactions

transaction: set of operations that occurs atomically

idea: something higher-level handles locking, etc.:

```
BeginTransaction();  
int FromOldBalance = GetBalance(FromAccount);  
int ToOldBalance = GetBalance(ToAccount);  
SetBalance(FromAccount, FromOldBalance - 100);  
SetBalance(ToAccount, ToOldBalance + 100);  
EndTransaction();
```

idea: library/database/etc. makes “transaction” happens all at once

consistency / durability

“happens all at once” = could mean:

locking to make sure no other operations interfere (consistency)

making sure on crash, no partial transaction seen (durability)

(some systems provide both, some provide only one)

we'll just talk about implementing consistency

implementing consistency: simple

simplest idea: only one run transaction at a time

implementing consistency: locking

everytime something read/written: acquire associated lock

on end transaction: release lock

if deadlock: undo everything, go back to BeginTransaction(), retry

how to undo?

one idea: keep list of writes instead of writing

apply writes only at EndTransaction()

implementing consistency: locking

everytime something read/written: acquire associated lock

on end transaction: release lock

if deadlock: **undo everything**, go back to BeginTransaction(), retry

how to undo?

one idea: keep list of writes instead of writing

apply writes only at EndTransaction()

implementing consistency: optimistic

on read: copy version # for value read

on write: record value to be written, but don't write yet

on end transaction:

- acquire locks on everything

- make sure values read haven't been changed since read

if they have changed, just retry transaction

recall: sockets

open connection then ...

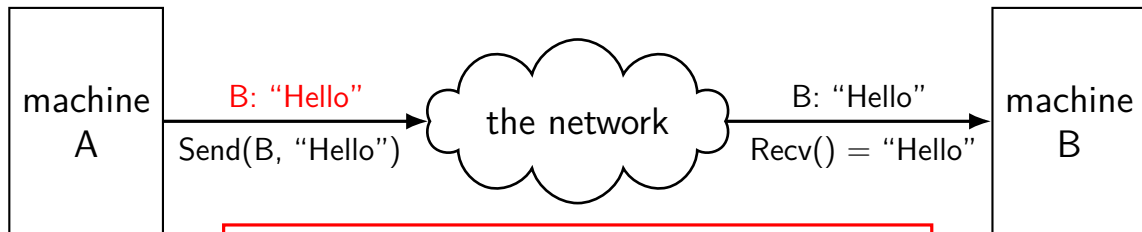
read+write just like a terminal file

doesn't look like individual messages

“connection abstraction”

mailbox model

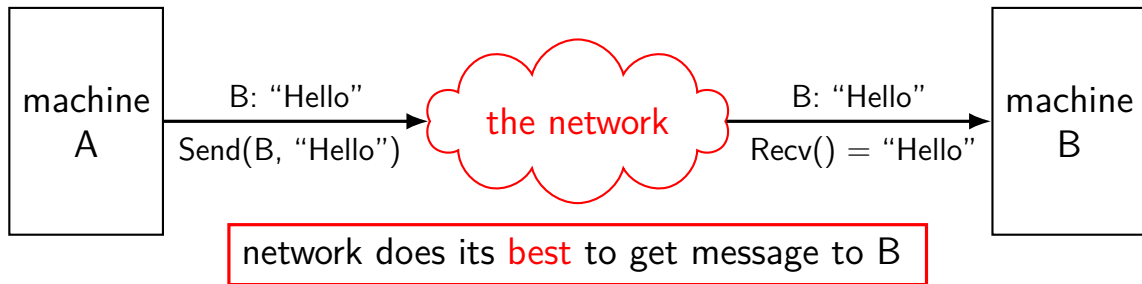
mailbox abstraction: send/receive messages



A sends "letter" to B
"envelope" tells network it's addressed to B
data in this example: "Hello"

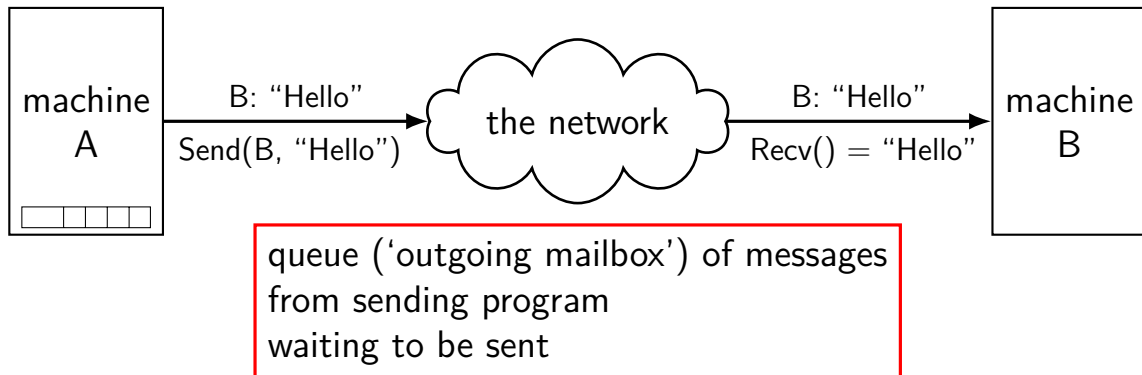
mailbox model

mailbox abstraction: send/receive messages



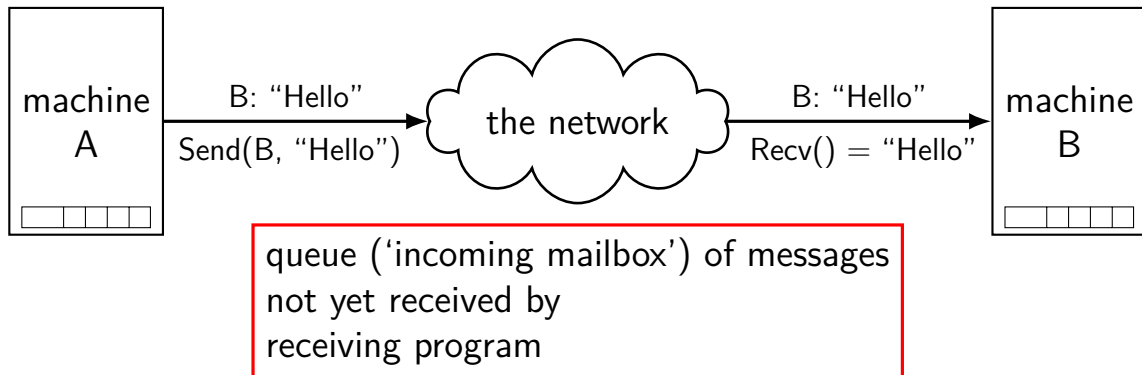
mailbox model

mailbox abstraction: send/receive messages



mailbox model

mailbox abstraction: send/receive messages



connections over mailboxes

real Internet: mailbox-style communication

send “letters” (packets) to particular mailboxes

have “envelope” (header) saying where they go

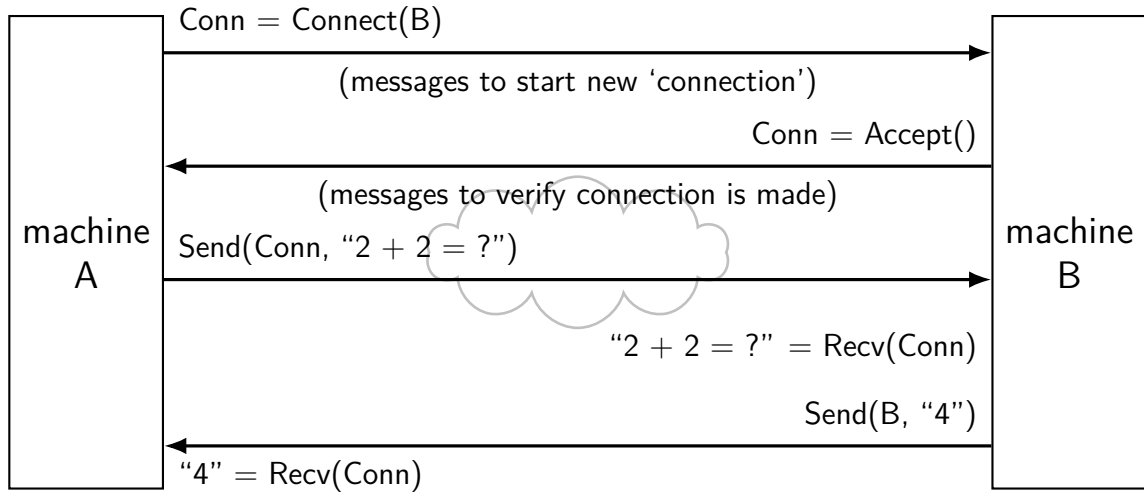
“best-effort”

no guarantee on order, when received

no guarantee on *if* received

sockets implemented on top of this

connections



layers

application	HTTP, SSH, SMTP, ...	application-defined meanings
transport	TCP, UDP, ...	reach correct program, reliability/streams
network	IPv4, IPv6, ...	reach correct machine (across networks)
link	Ethernet, Wi-Fi, ...	coordinate shared wire/radio
physical	...	encode bits for wire/radio

layers terminology

application	application-defined meanings	
transport	reach correct program, reliability/streams	segments/datagrams
network	reach correct machine (across networks)	packets
link	coordinate shared wire/radio	frames
physical	encode bits for wire/radio	

layer wrapping

upper layers usually implemented using lower layers

example: implement reliable + large messages (transport layer)
by sending multiple unreliable messages across networks (network layer)

example: implement reaching machine across networks (network layer)
by sending multiple messages on local networks (link layer)

layers

application	HTTP, SSH, SMTP, ...	application-defined meanings
transport	TCP, UDP, ...	reach correct program, reliability/streams
network	IPv4, IPv6, ...	reach correct machine (across networks)
link	Ethernet, Wi-Fi, ...	coordinate shared wire/radio
physical	...	encode bits for wire/radio

network limitations/failures

messages lost

messages delayed/reordered

messages limited in size

messages corrupted

network limitations/failures

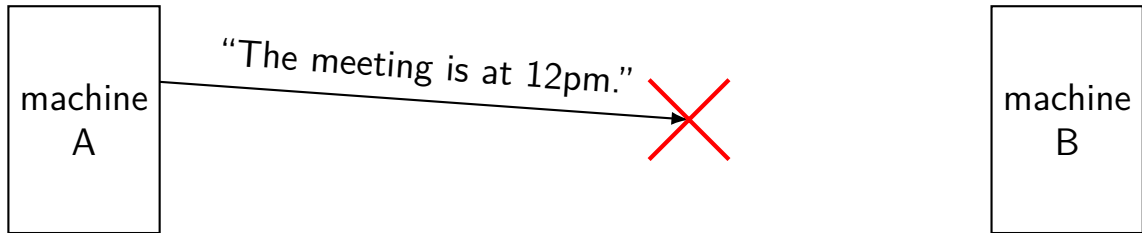
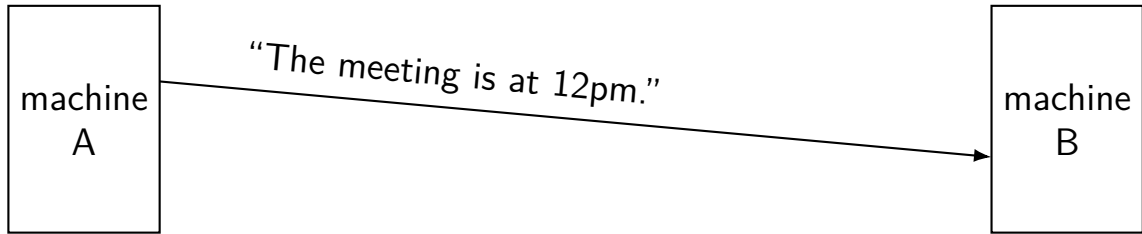
messages lost

messages delayed/reordered

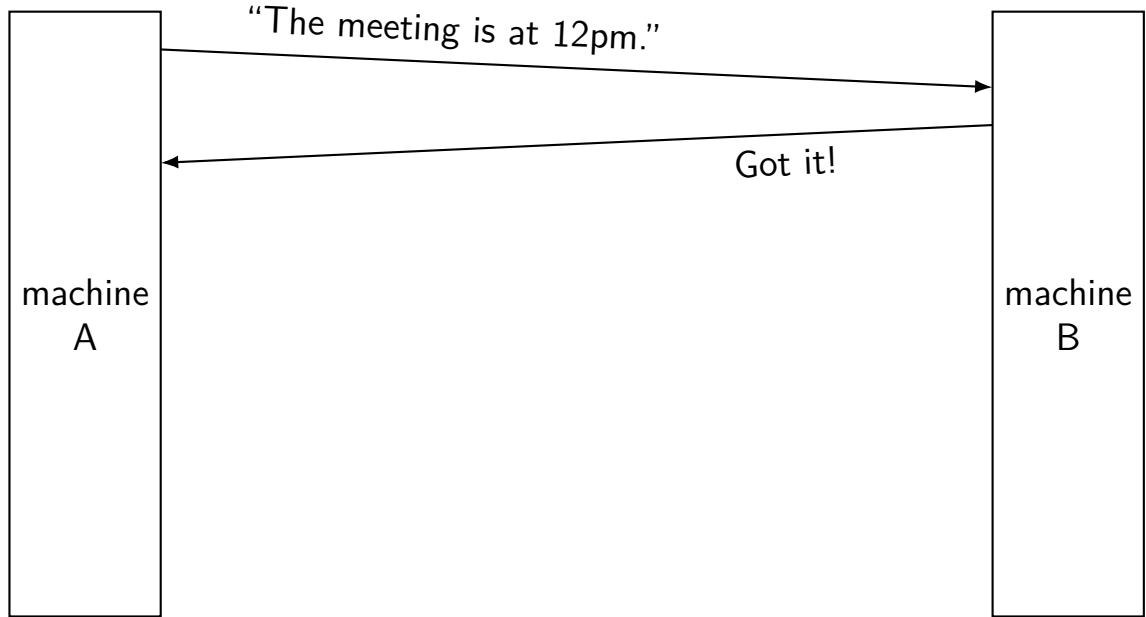
messages limited in size

messages corrupted

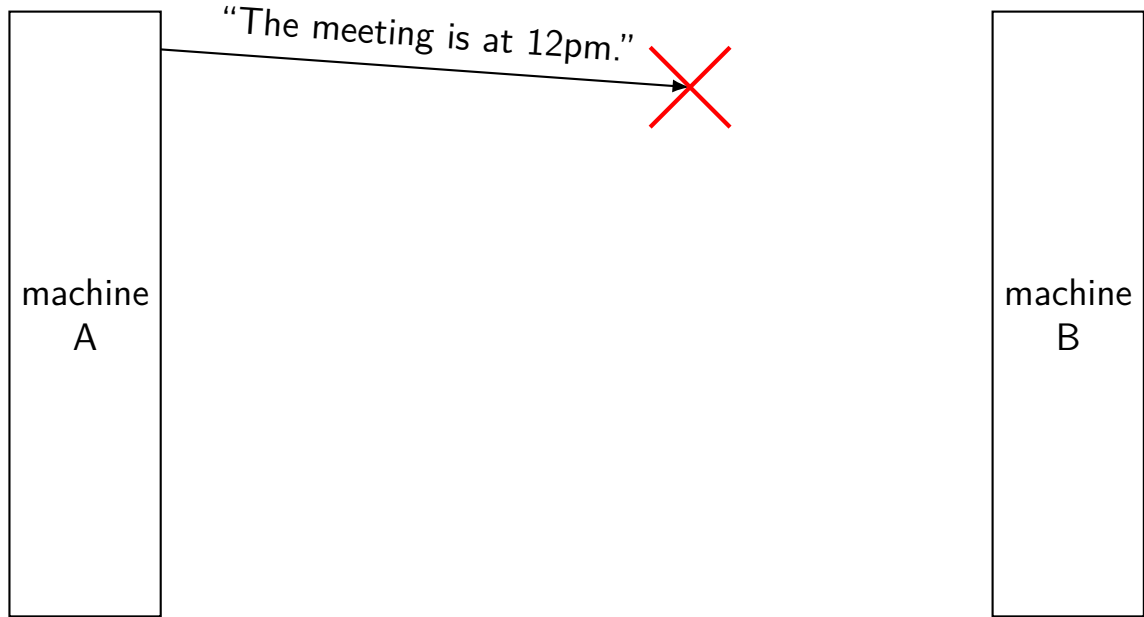
dealing with network message lost



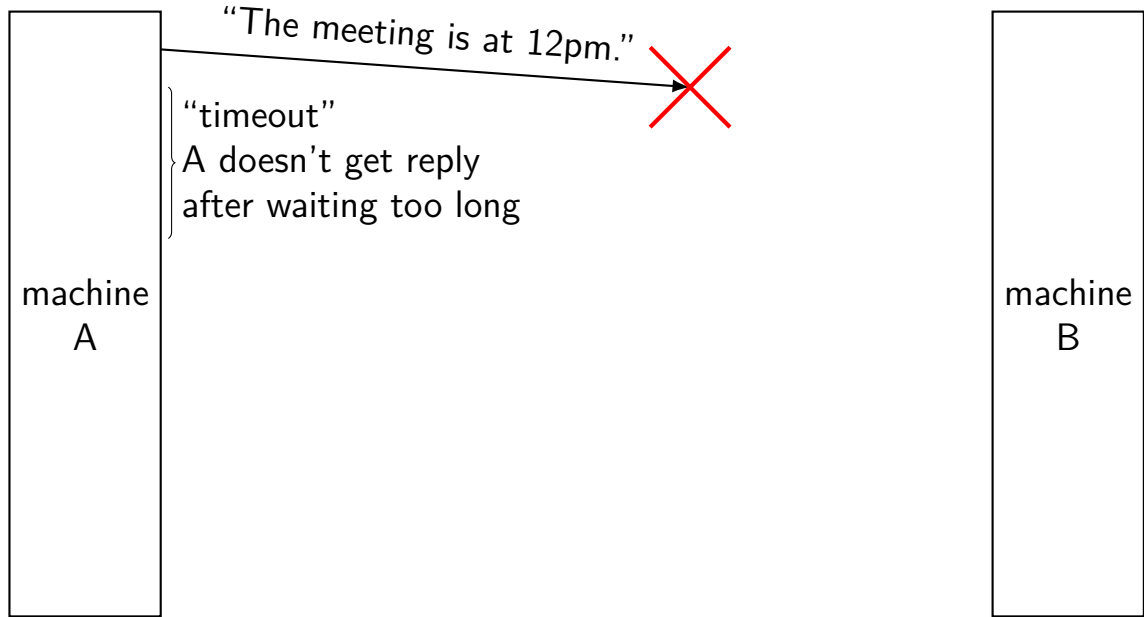
handling lost message: acknowledgements



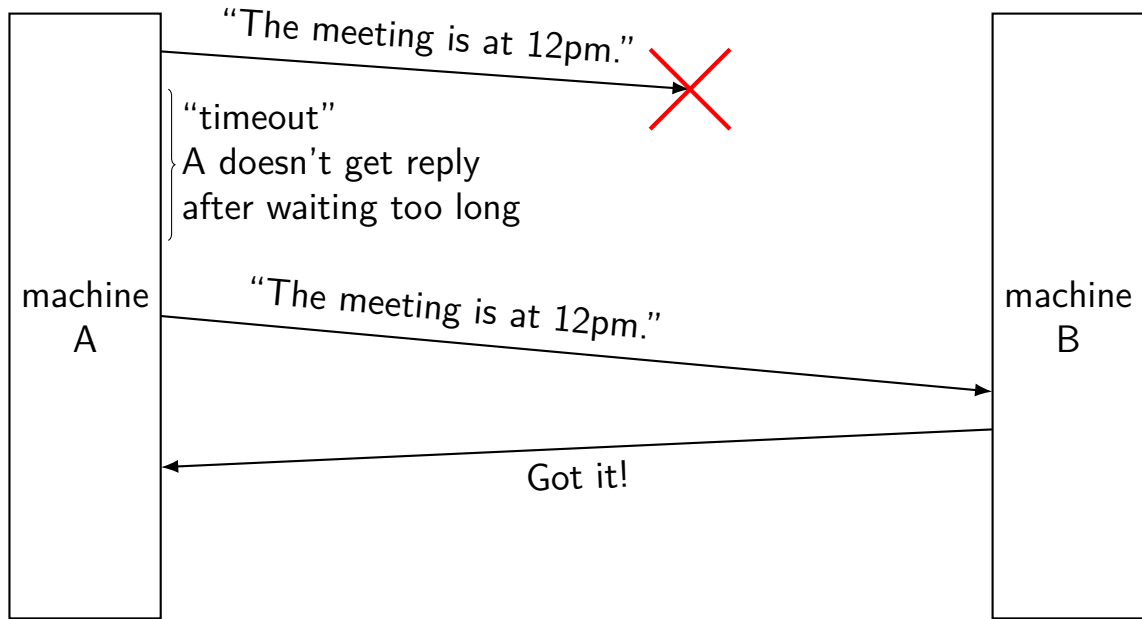
handling lost message



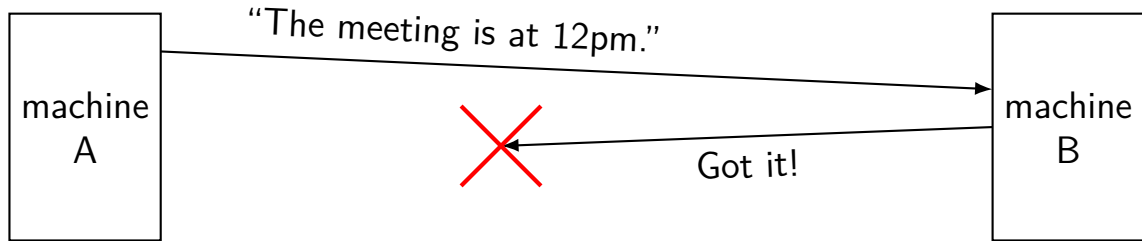
handling lost message



handling lost message



exercise: lost acknowledgement



exercise: how to fix this?

- A. machine A needs to send "Got 'got it!' "
- B. machine B should resend "Got it!" on its own
- C. machine A should resend the original message on its own
- D. none of these

network limitations/failures

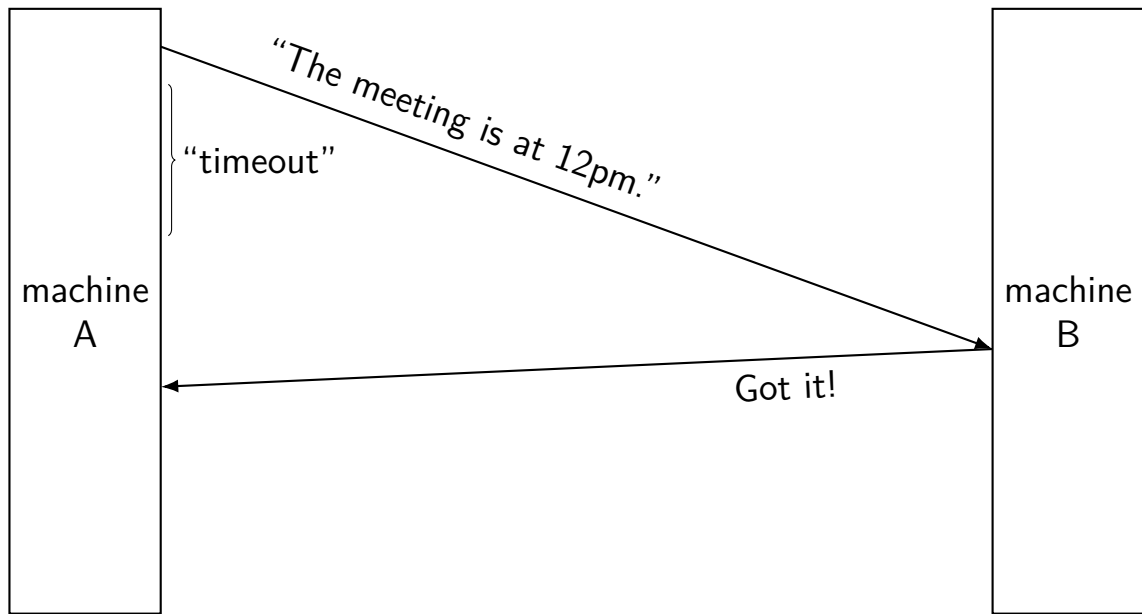
messages lost

messages delayed/reordered

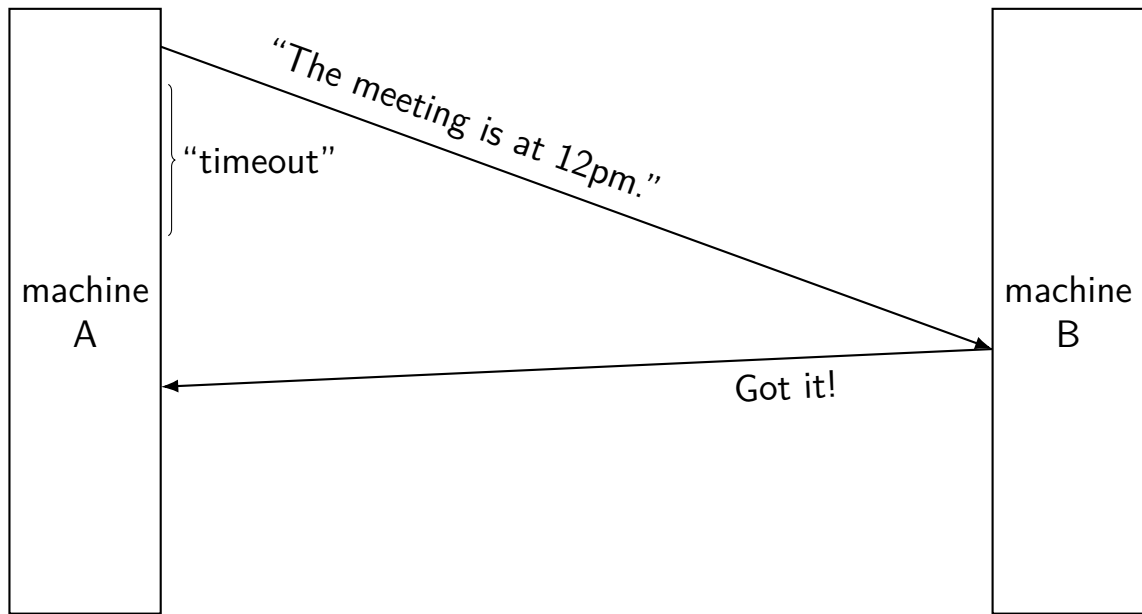
messages limited in size

messages corrupted

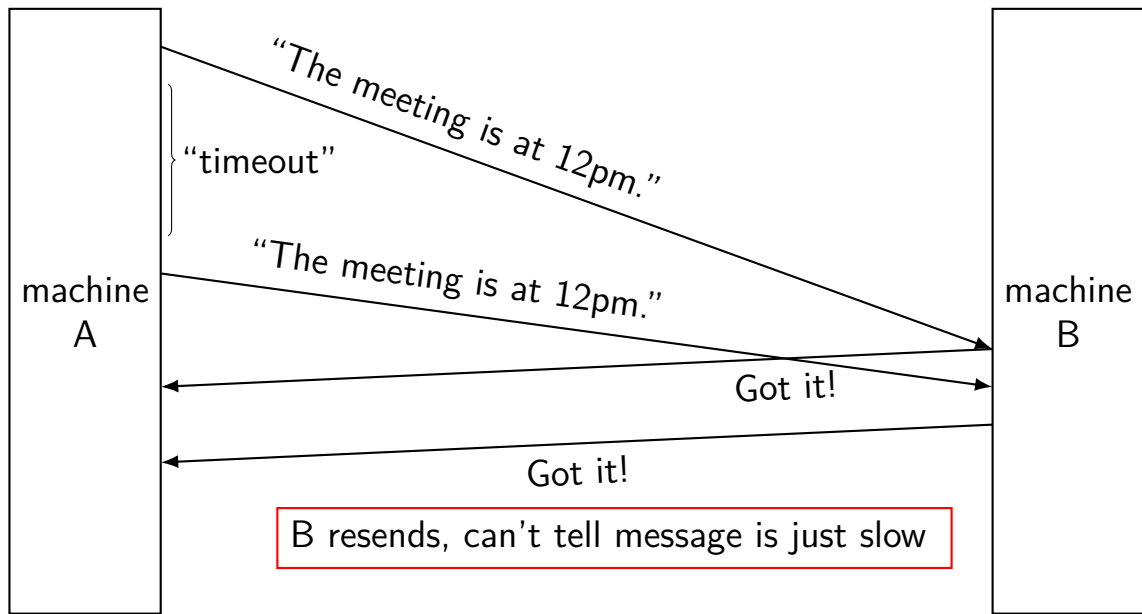
delayed message



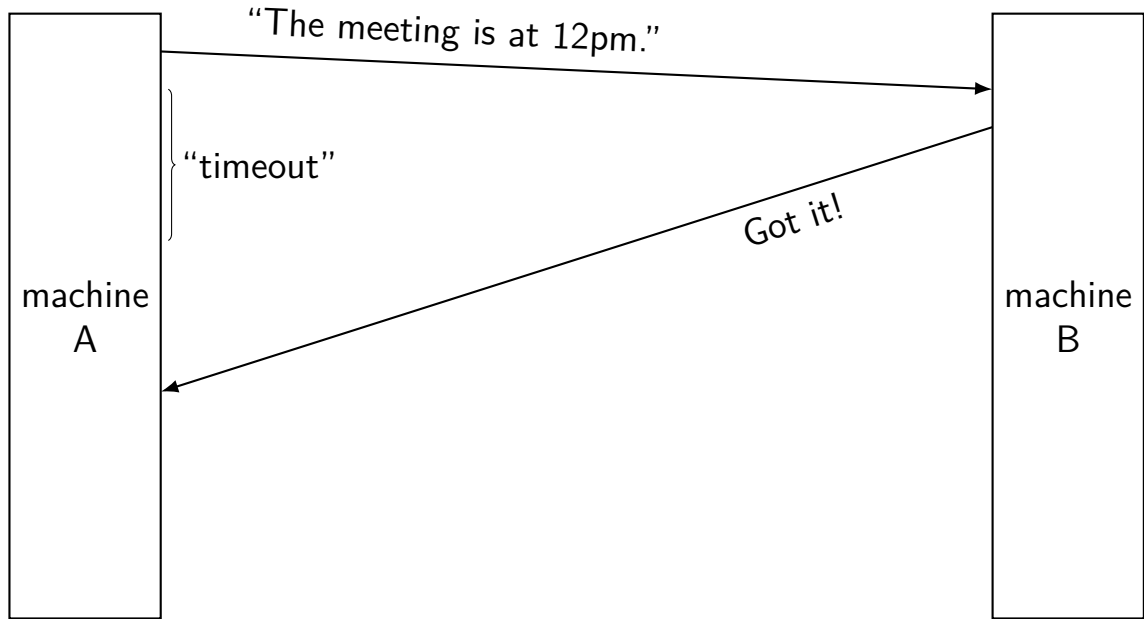
delayed message



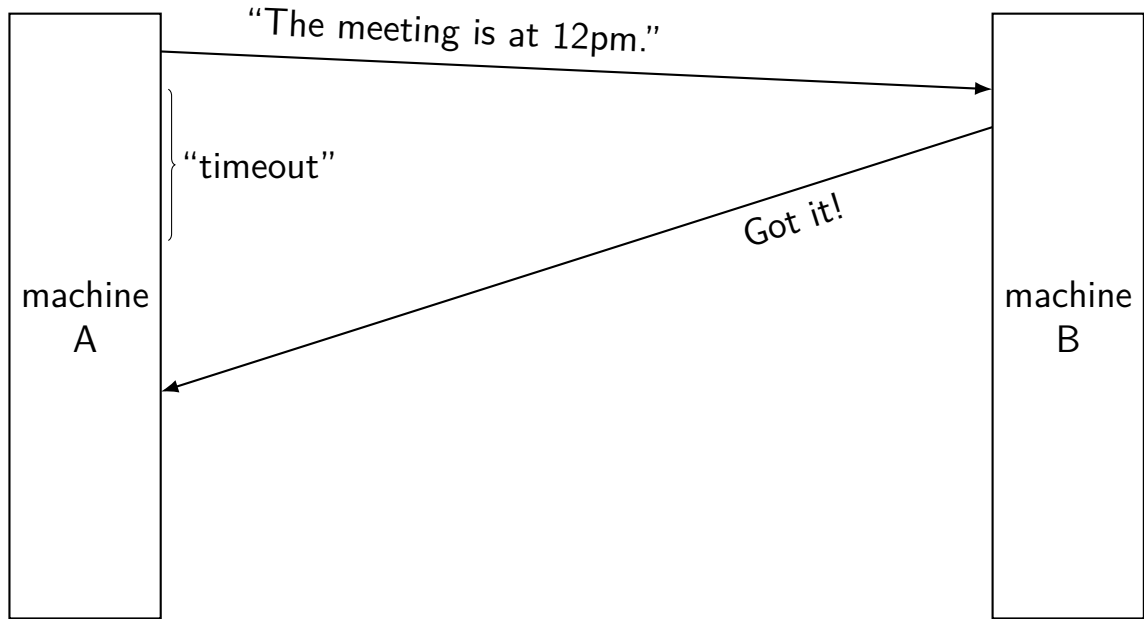
delayed message



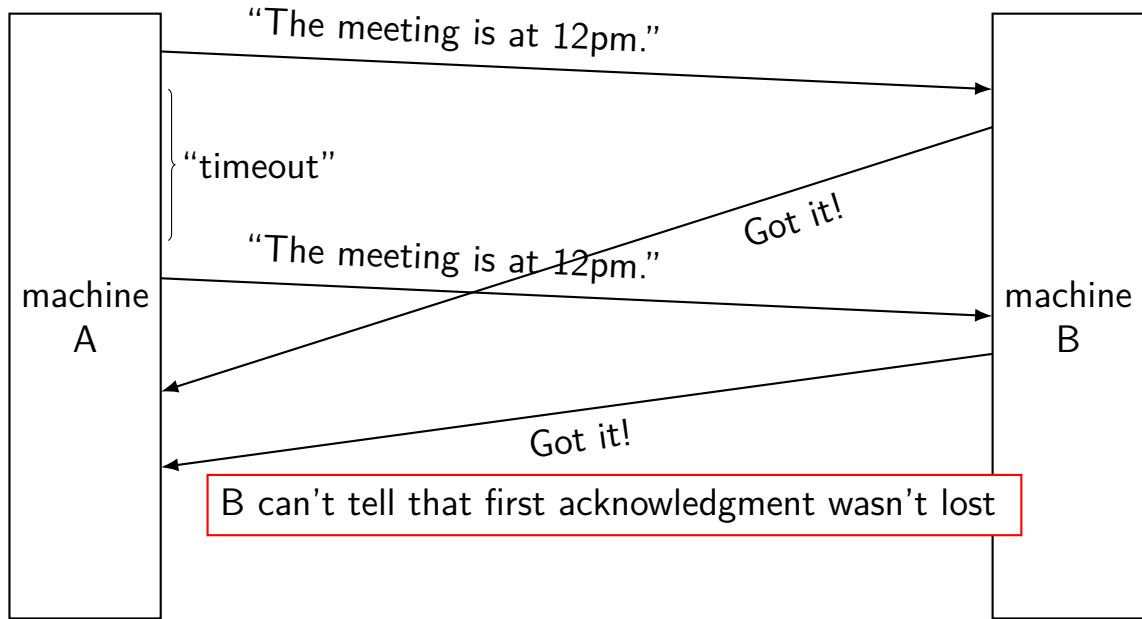
delayed acknowledgements



delayed acknowledgements



delayed acknowledgements



network limitations/failures

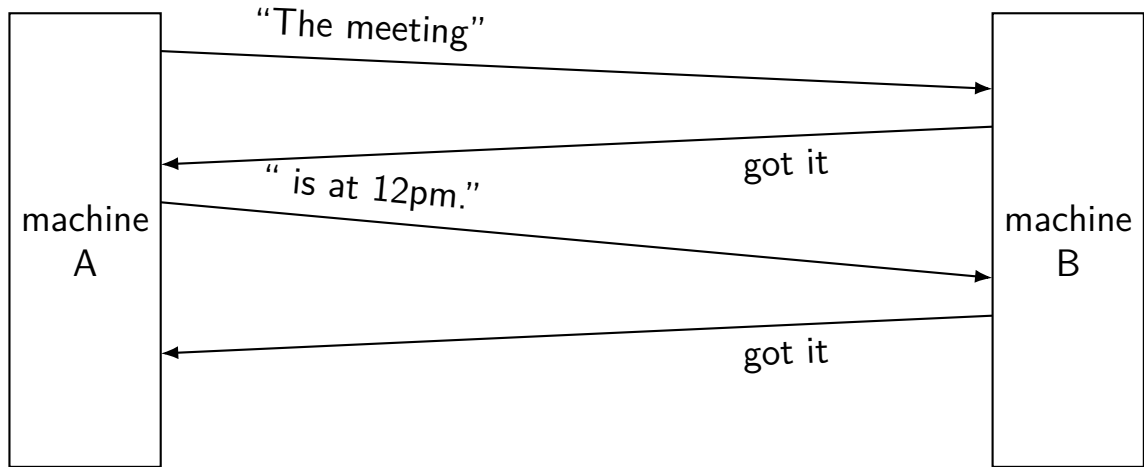
messages lost

messages delayed/reordered

messages limited in size

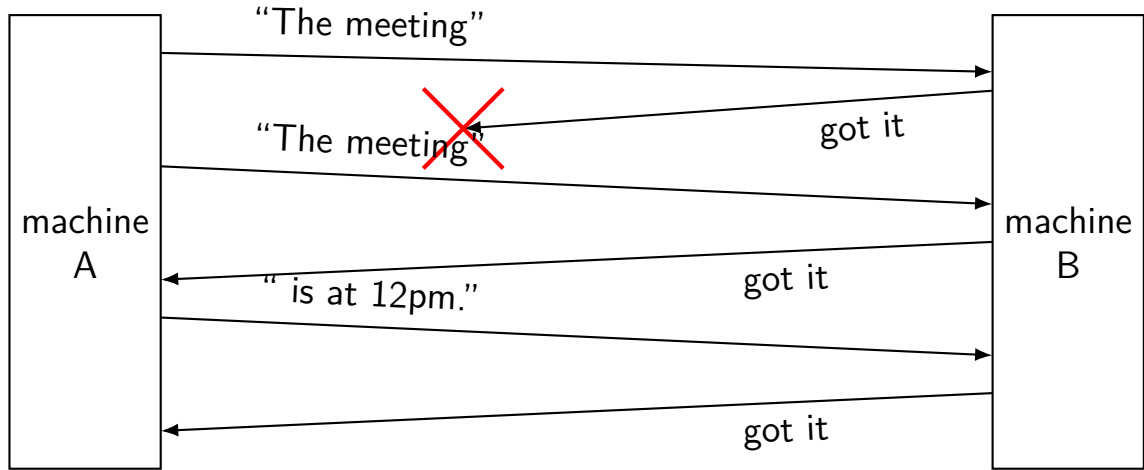
messages corrupted

splitting messages: try 1

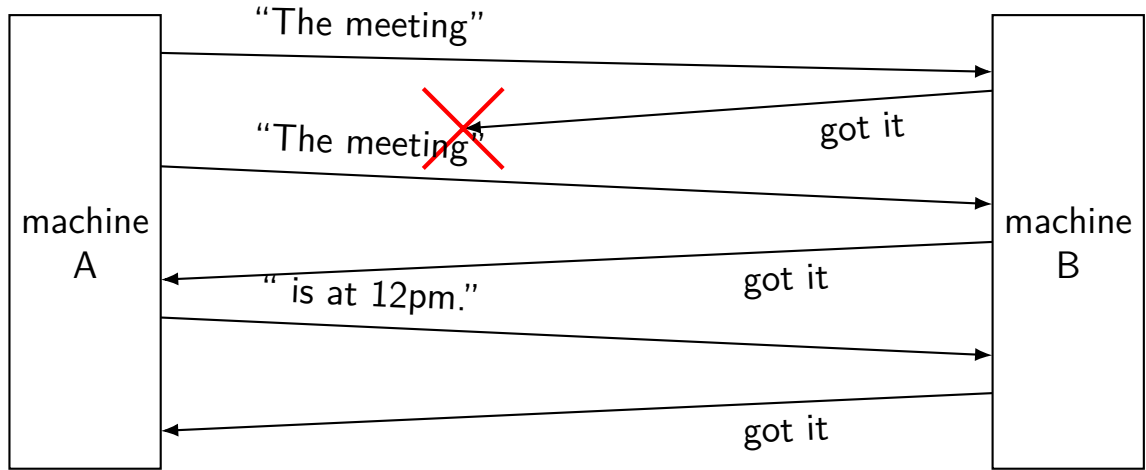


reconstructed message:
The meeting is at 12pm.

splitting messages: try 1 — problem 1



splitting messages: try 1 — problem 1



reconstructed message:

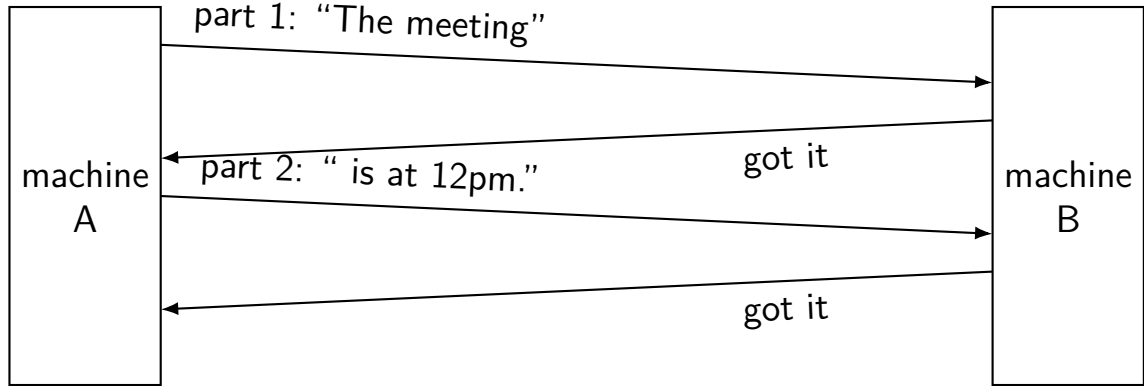
The meetingThe meeting is at 12pm.

exercise: other problems?

other scenarios where we'd also have problems?

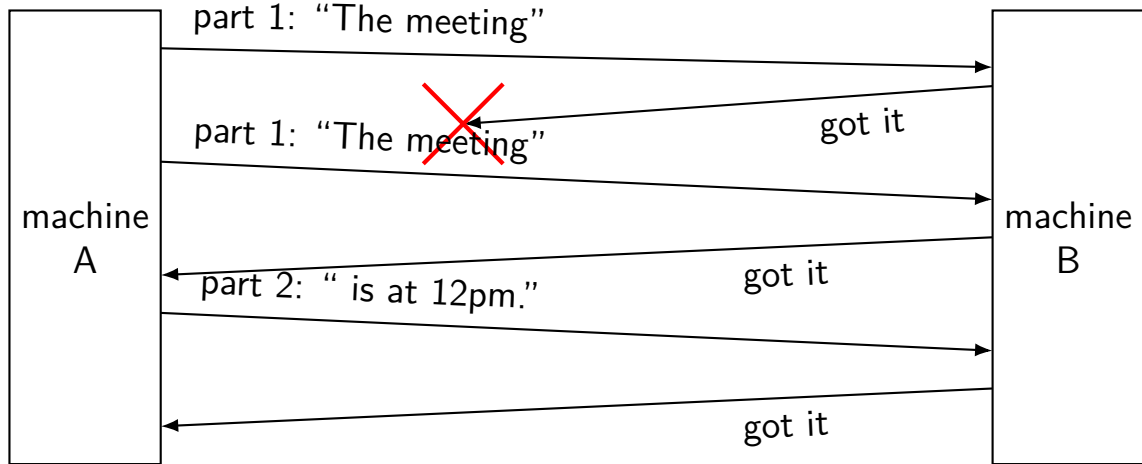
1. message (instead of acknowledgment) is lost
2. first message from machine A is delayed a long time by network
3. acknowledgment of second message lost instead of first

splitting messages: try 2



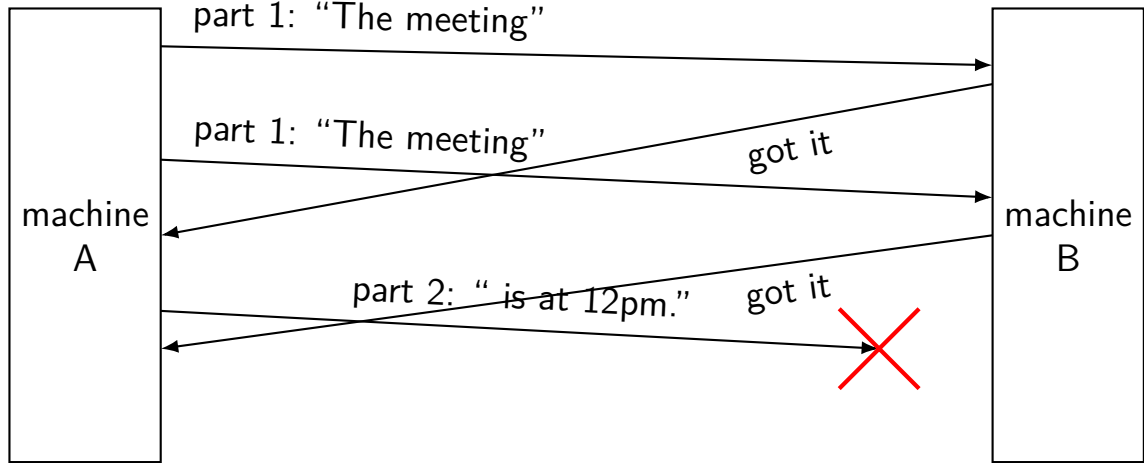
reconstructed message:
The meeting is at 12pm.

splitting messages: try 2 — missed ack



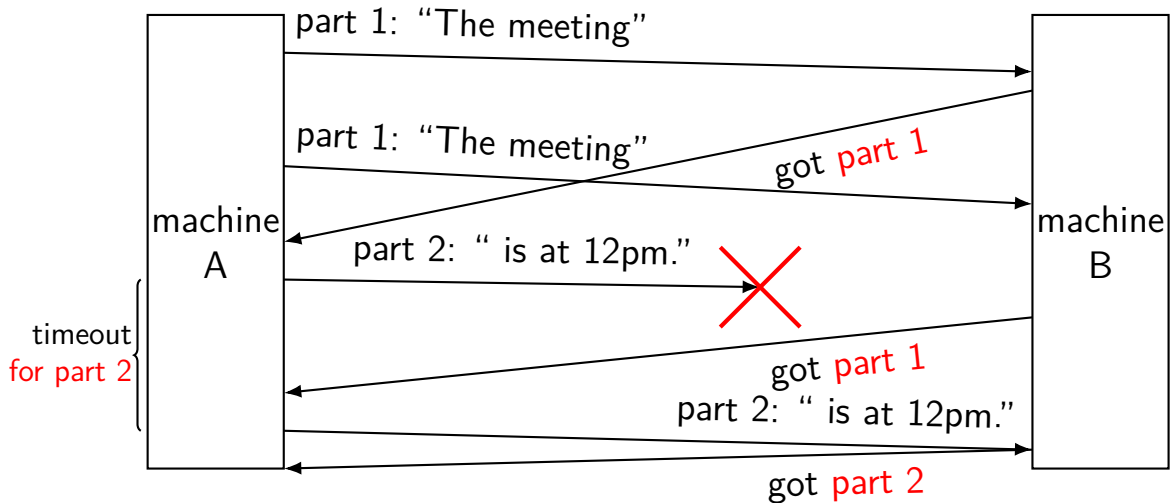
reconstructed message:
The meeting is at 12pm.

splitting messages: try 2 — problem



A thinks: part 1 + part 2 acknowledged!

splitting messages: version 3



network limitations/failures

messages lost

messages delayed/reordered

messages limited in size

messages corrupted

message corrupted

instead of sending “message”

say $\text{Hash}(\text{“message”}) = 0x\text{ABCDEF12}$

then send “0xABCDEF12,message”

when receiving, recompute hash

pretend message lost if does not match

“checksum”

these hashes commonly called “checksums”

in UDP/TCP, hash function: treat bytes of messages as array of integers; then add integers together

going faster

so far: send one message, get acknowledgments

pretty slow

instead, can send a bunch of parts and get them acknowledged together

need to do *congestion control* to avoid overloading network

layers

application	HTTP, SSH, SMTP, ...	application-defined meanings
transport	TCP, UDP, ...	reach correct program, reliability/streams
network	IPv4, IPv6, ...	reach correct machine (across networks)
link	Ethernet, Wi-Fi, ...	coordinate shared wire/radio
physical	...	encode bits for wire/radio

more than four layers?

sometimes more layers above 'application'

e.g. HTTPS:

HTTP (app layer) on TLS (another app layer) on TCP (network) on ...

e.g. DNS over HTTPS:

DNS (app layer) on HTTP on on TLS on TCP on ...

e.g. SFTP:

SFTP (app layer??) on SSH (another app layer) on TCP on ...

e.g. HTTP over OpenVPN:

HTTP on TCP on IP on OpenVPN on UDP on different IP on ...

names and addresses

name	address
logical identifier	location/how to locate
variable counter	memory address 0x7FFF9430
DNS name www.virginia.edu	IPv4 address 128.143.22.36
DNS name mail.google.com	IPv4 address 216.58.217.69
DNS name mail.google.com	IPv6 address 2607:f8b0:4004:80b::2005
DNS name reiss-t3620.cs.virginia.edu	IPv4 address 128.143.67.91
DNS name reiss-t3620.cs.virginia.edu	MAC address 18:66:da:2e:7f:da
service name https	port number 443
service name ssh	port number 22

layers

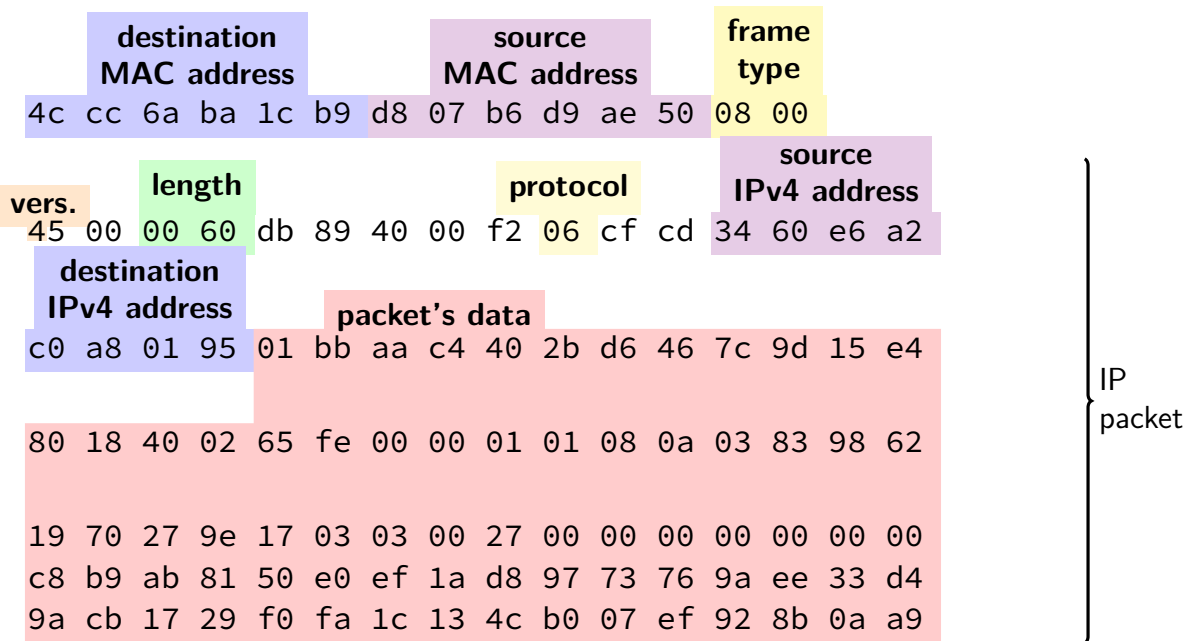
application	HTTP, SSH, SMTP, ...	application-defined meanings
transport	TCP, UDP, ...	reach correct program, reliability/streams
network	IPv4, IPv6, ...	reach correct machine (across networks)
link	Ethernet, Wi-Fi, ...	coordinate shared wire/radio
physical	...	encode bits for wire/radio

an Ethernet frame

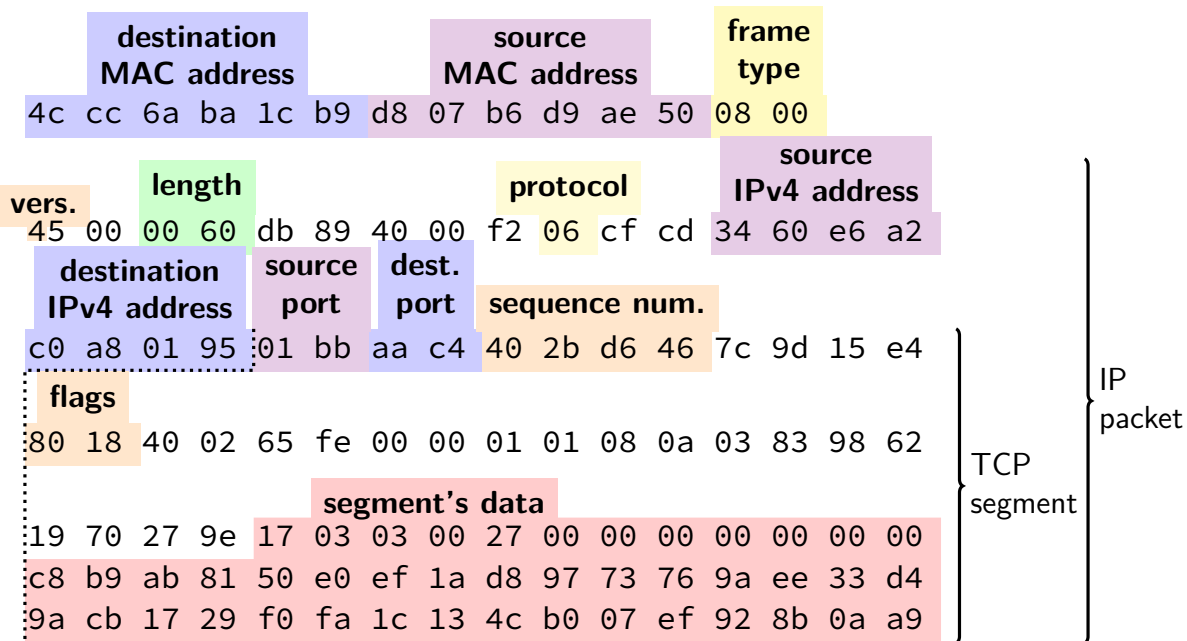
destination MAC address						source MAC address						frame type	
4c	cc	6a	ba	1c	b9	d8	07	b6	d9	ae	50	08	00

frame's data															
45	00	00	60	db	89	40	00	f2	06	cf	cd	34	60	e6	a2
c0	a8	01	95	01	bb	aa	c4	40	2b	d6	46	7c	9d	15	e4
80	18	40	02	65	fe	00	00	01	01	08	0a	03	83	98	62
19	70	27	9e	17	03	03	00	27	00	00	00	00	00	00	00
c8	b9	ab	81	50	e0	ef	1a	d8	97	73	76	9a	ee	33	d4
9a	cb	17	29	f0	fa	1c	13	4c	b0	07	ef	92	8b	0a	a9

an Ethernet frame



an Ethernet frame



layers

application	HTTP, SSH, SMTP, ...	application-defined meanings
transport	TCP, UDP, ...	reach correct program, reliability/streams
network	IPv4, IPv6, ...	reach correct machine (across networks)
link	Ethernet, Wi-Fi, ...	coordinate shared wire/radio
physical	...	encode bits for wire/radio

the network layer

the Internet Protocol (IP) version 4 or version 6

there are also others, but quite uncommon today

allows send messages to/recv messages from other networks

“internetwork”

messages usually called “packets”

IPv4 addresses

32-bit numbers

typically written like 128.143.67.11

four 8-bit decimal values separated by dots

first part is most significant

same as $128 \cdot 256^3 + 143 \cdot 256^2 + 67 \cdot 256 + 11 = 2\,156\,782\,459$

organizations get blocks of IPs

e.g. UVA has 128.143.0.0–128.143.255.255

e.g. Google has 216.58.192.0–216.58.223.255 and

74.125.0.0–74.125.255.255 and 35.192.0.0–35.207.255.255

some IPs reserved for non-Internet use (127.*, 10.*, 192.168.*)

IPv6 addresses

IPv6 like IPv4, but with 128-bit numbers

written in hex, 16-bit parts, separated by colons (:)

strings of 0s represented by double-colons (::)

typically given to users in blocks of 2^{80} or 2^{64} addresses
no need for address translation?

2607:f8b0:400d:c00::6a =

2607:f8b0:400d:0c00:0000:0000:0000:006a

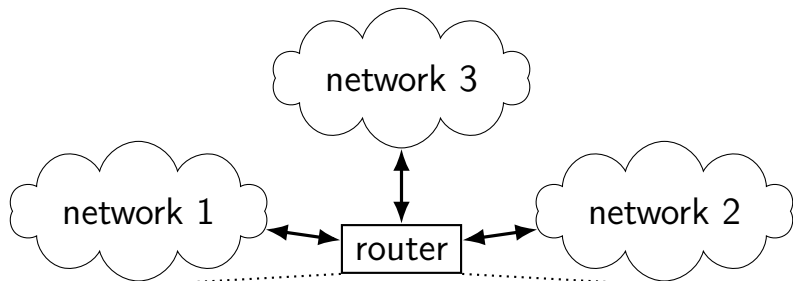
2607f8b0400d0c0000000000000000006a_{SIXTEEN}

selected special IPv6 addresses

`::1` = localhost

anything starting with `fe80` = link-local addresses
never forwarded by routers

IPv4 addresses and routing tables



if I receive data for...	send it to...
128.143.0.0—128.143.255.255	network 1, 11.4.3.2
192.107.102.0—192.107.102.255	network 1, 11.4.3.2
...	...
4.0.0.0—7.255.255.255	network 2, 12.4.6.4
64.8.0.0—64.15.255.255	network 2, 45.4.0.1
...	...
anything else	network 3, 199.44.33.1

selected special IPv4 addresses

127.0.0.0 — 127.255.255.255 — localhost

AKA loopback

the machine we're on

typically only 127.0.0.1 is used

192.168.0.0–192.168.255.255 and

10.0.0.0–10.255.255.255 and

172.16.0.0–172.31.255.255

“private” IP addresses

not used on the Internet

commonly connected to Internet with **network address translation**

also 100.64.0.0–100.127.255.255 (but with restrictions)

169.254.0.0–169.254.255.255

link-local addresses — ‘never’ forwarded by routers

layers

application	HTTP, SSH, SMTP, ...	application-defined meanings
transport	TCP, UDP, ...	reach correct program, reliability/streams
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physical	...	encode bits for wire/radio

port numbers

we run multiple programs on a machine

IP addresses identifying machine — not enough

port numbers

we run multiple programs on a machine

IP addresses identifying machine — not enough

so, add 16-bit *port numbers*

think: multiple PO boxes at address

port numbers

we run multiple programs on a machine

IP addresses identifying machine — not enough

so, add 16-bit *port numbers*

think: multiple PO boxes at address

0–49151: typically assigned for particular services

80 = http, 443 = https, 22 = ssh, ...

49152–65535: allocated on demand

default “return address” for client connecting to server

UDP v TCP

TCP: stream to other program

- reliable transmission of as much data as you want

- “connecting” fails if server not responding

- `write(fd, "a", 1); write(fd, "b", 1) = write(fd, "ab", 2)`

- (at least) one socket per remote program being talked to

UDP: messages sent to program, but no reliability/streams

- unreliable transmission of short messages

- `write(fd, "a", 1); write(fd, "b", 1)  $\neq$  write(fd, "ab", 2)`

- “connecting” just sets default destination

- can `sendto()/recvfrom()` multiple other programs with one socket

- (but don't have to)

UDP v TCP

TCP: stream to other program

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“connecting” just sets default destination

can `sendto()/recvfrom()` multiple other programs with one socket

(but don't have to)

connections in TCP/IP

connection identified by *5-tuple*

used by OS to lookup “where is the socket?”

(protocol=TCP/UDP, local IP addr., local port, remote IP addr., remote port)

local IP address, port number can be set with `bind()` function

typically always done for servers, not done for clients

system will choose default if you don't

connections on my desktop

```
cr4bd@reiss-t3620>/u/cr4bd
```

```
$ netstat —inet —inet6 —numeric
```

```
Active Internet connections (w/o servers)
```

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	128.143.67.91:49202	128.143.63.34:22	ESTABLISH
tcp	0	0	128.143.67.91:803	128.143.67.236:2049	ESTABLISH
tcp	0	0	128.143.67.91:50292	128.143.67.226:22	TIME_WAIT
tcp	0	0	128.143.67.91:54722	128.143.67.236:2049	TIME_WAIT
tcp	0	0	128.143.67.91:52002	128.143.67.236:111	TIME_WAIT
tcp	0	0	128.143.67.91:732	128.143.67.236:63439	TIME_WAIT
tcp	0	0	128.143.67.91:40664	128.143.67.236:2049	TIME_WAIT
tcp	0	0	128.143.67.91:54098	128.143.67.236:111	TIME_WAIT
tcp	0	0	128.143.67.91:49302	128.143.67.236:63439	TIME_WAIT
tcp	0	0	128.143.67.91:50236	128.143.67.236:111	TIME_WAIT
tcp	0	0	128.143.67.91:22	172.27.98.20:49566	ESTABLISH
tcp	0	0	128.143.67.91:51000	128.143.67.236:111	TIME_WAIT
tcp	0	0	127.0.0.1:50438	127.0.0.1:631	ESTABLISH
tcp	0	0	127.0.0.1:631	127.0.0.1:50438	ESTABLISH

non-connection sockets

TCP servers waiting for connections +
UDP sockets with no particular remote host

Linux: OS keeps 5-tuple with “wildcard” remote address

“listening” sockets on my desktop

```
cr4bd@reiss-t3620>/u/cr4bd
```

```
$ netstat —inet —inet6 —numeric —listen
```

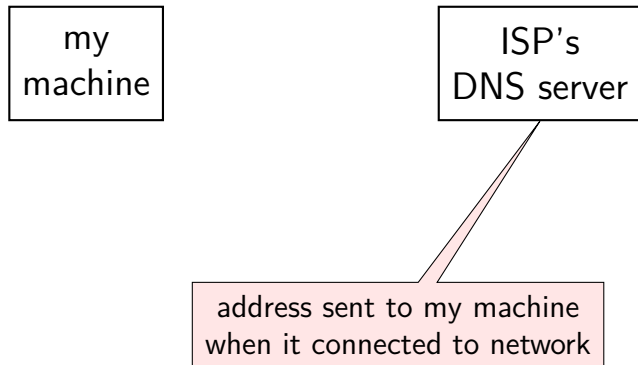
```
Active Internet connections (only servers)
```

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	127.0.0.1:38537	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.1:36777	0.0.0.0:*	LISTEN
tcp	0	0	0.0.0.0:41099	0.0.0.0:*	LISTEN
tcp	0	0	0.0.0.0:45291	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.1:51949	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.1:41071	0.0.0.0:*	LISTEN
tcp	0	0	0.0.0.0:111	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.1:32881	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.1:38673	0.0.0.0:*	LISTEN
...					
tcp6	0	0	:::42689	:::*	LISTEN
udp	0	0	128.143.67.91:60001	0.0.0.0:*	
udp	0	0	128.143.67.91:60002	0.0.0.0:*	
...					
udp6	0	0	:::59938	:::*	

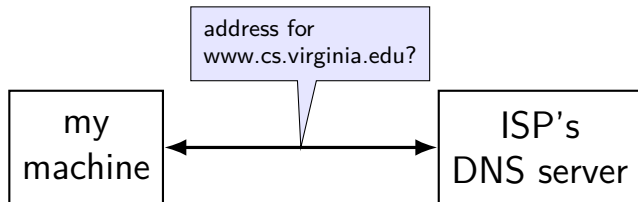
names and addresses

name	address
logical identifier	location/how to locate
variable counter	memory address 0x7FFF9430
DNS name www.virginia.edu	IPv4 address 128.143.22.36
DNS name mail.google.com	IPv4 address 216.58.217.69
DNS name mail.google.com	IPv6 address 2607:f8b0:4004:80b::2005
DNS name reiss-t3620.cs.virginia.edu	IPv4 address 128.143.67.91
DNS name reiss-t3620.cs.virginia.edu	MAC address 18:66:da:2e:7f:da
service name https	port number 443
service name ssh	port number 22

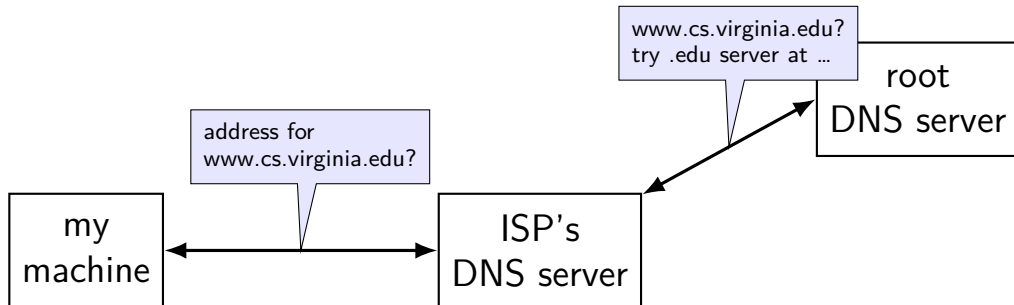
DNS: distributed database



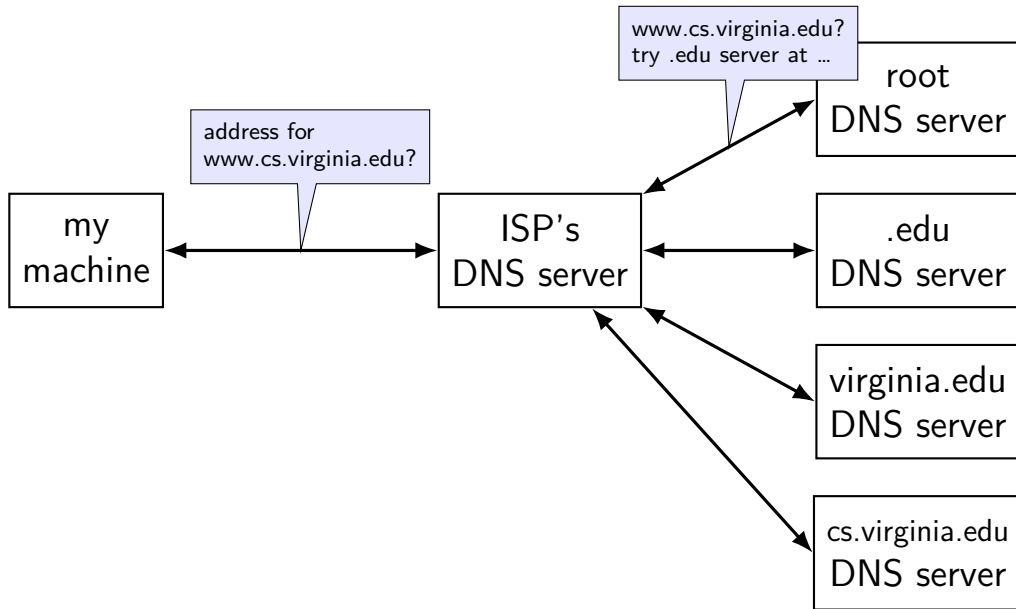
DNS: distributed database



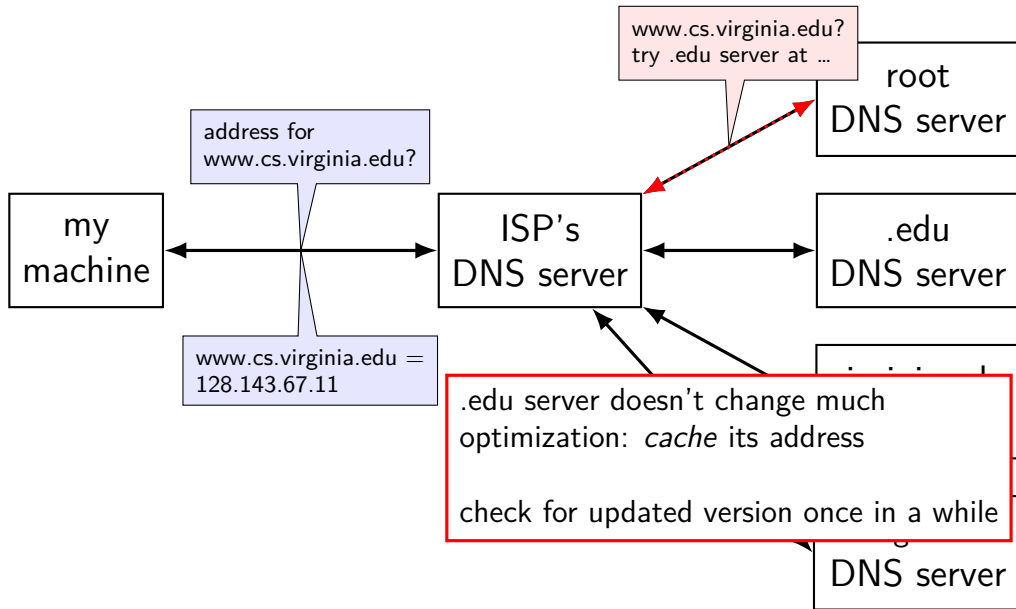
DNS: distributed database



DNS: distributed database



DNS: distributed database



URL / URIs

Uniform Resource Locators (URL)

tells how to find “resource” on network

uniform — one syntax for multiple protocols (types of servers, etc.)

Uniform Resource Identifiers

superset of URLs

URI examples

`https://kytos02.cs.virginia.edu:443/cs3130-spring2023/
quizzes/quiz.php?qid=02#q2`

`https://kytos02.cs.virginia.edu/cs3130-spring2023/
quizzes/quiz.php?qid=02`

`https://www.cs.virginia.edu/`

`sftp://cr4bd@portal.cs.virginia.edu/u/cr4bd/file.txt`

`tel:+1-434-982-2200`

`//www.cs.virginia.edu/~cr4bd/3130/S2023/
/~cr4bd/3130/S2023`

scheme and/or host implied from context

URI generally

scheme://authority/path?query#fragment

scheme: — what protocol

//authority/

authority = user@host:port OR host:port OR user@host OR host

path

which resource

?query — usually key/value pairs

#fragment — place in resource

most components (sometimes) optional

autoconfiguration

problem: how does my machine get IP address

otherwise:

- have sysadmin type one in?

- just choose one?

- ask machine on local network to assign it

autoconfiguration

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- have sysadmin type one in?

- just choose one?

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autoconfiguration

problem: how does my machine get IP address

otherwise:

- have sysadmin type one in?

- just choose one?

- ask machine on local network to assign it

often local router machine runs service to assign IP addresses

- knows what IP addresses are available

- sysadmin might configure in mapping from MAC addresses to IP addresses

DHCP high-level

protocol done over UDP

but since we don't have IP address yet, use 0.0.0.0

and since we don't know server address, use 255.255.255.255
= “everyone on the local network”

local server replies to request with address + time limit

later: can send messages to local server to renew/give up address

DHCP high-level

protocol done over UDP

but since we don't have IP address yet, use 0.0.0.0

and since we don't know server address, use 255.255.255.255
= “everyone on the local network”

local server replies to request with address + **time limit**

later: can send messages to local server to renew/give up address

exercise: why time limit?

DHCP “lease”

rather than getting address forever

but DHCP has way of releasing taken address

why impose a time limit

network address translation

IPv4 addresses are kinda scarce

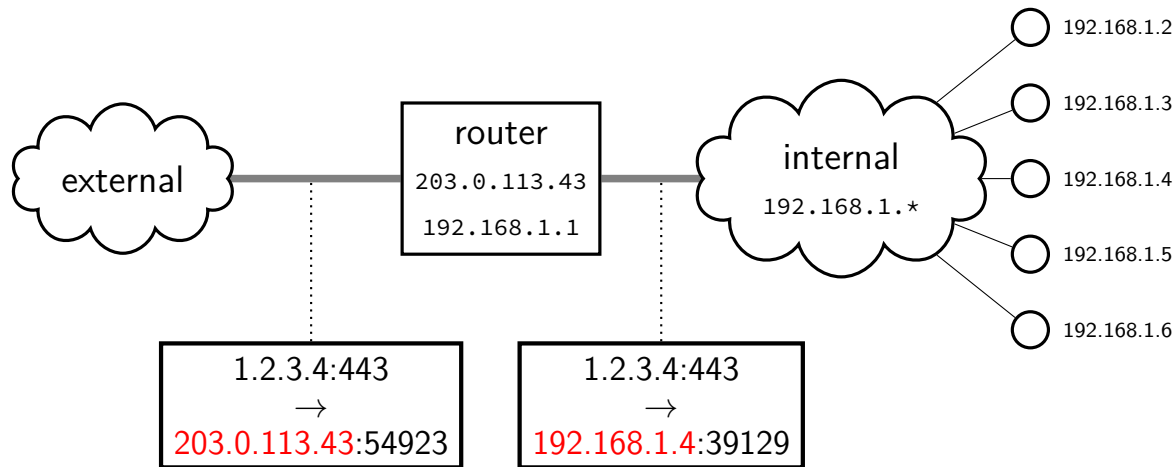
solution: *convert* many private addrs. to one public addr.

locally: use private IP addresses for local addresses

outside: private IP addresses become a single public one

commonly how home networks work (and some ISPs)

NAT idea



NAT illusion

NAT illusion:

private IP address communicating directly with public IP

inside network, talking to outside:

- use private local address
- use public remote address
- never see router's address

outside network, talking to inside

- use public local address
- use router's public address

implementing NAT

remote host + port	outside local port number	inside IP	inside port number
128.148.17.3:443	54033	192.168.1.5	43222
11.7.17.3:443	53037	192.168.1.5	33212
128.148.31.2:22	54032	192.168.1.37	43010
128.148.17.3:443	63039	192.168.1.37	32132

table of the translations

need to update as new connections made

upcoming lab

request + receive message split into pieces

you are responsible for:

- requesting parts in order

- resending requests if messages lost/corrupted

“acknowledge” receiving part X to request part $X+1$

upcoming lab

request + receive message split into pieces

you are responsible for:

- requesting parts in order

- resending requests if messages lost/corrupted

“acknowledge” receiving part X to request part $X+1$

protocol

GET x — retrieve message x ($x = 0, 1, 2$, or 3)

- other end acknowledges by giving data

- if they don't reply, you need to send again

- higher numbered messages have errors/etc. that are harder to handle

ACK n

- request message $n + 1$ by acknowledging message n

- not quite same purpose as acknowledgments in prior examples

- (in lab, the response is your 'acknowledgment' of your request;

- you retry if you don't get it)

protocol

GET x — retrieve message x ($x = 0, 1, 2$, or 3)

other end acknowledges by giving data

if they don't reply, you need to send again

higher numbered messages have errors/etc. that are harder to handle

ACK n

request message $n + 1$ by acknowledging message n

not quite same purpose as acknowledgments in prior examples

(in lab, the response is your 'acknowledgment' of your request;
you retry if you don't get it)

callback-based programming (1)

```
/* library code you don't write */
/* lab: part of waitForAllTimeoutsAndMessagesThenExit() */
void mainLoop() {
    while (notExiting) {
        Event event = waitForAndGetNextEvent();
        if (event.type == RECIEVED) {
            recvd(...);
        } else if (event.type == TIMEOUT) {
            (event.timeout_function)(...);
        }
        ...
    }
}
```

callback-based programming (2)

```
/* your code, called by library */
void recvd(...) {
    ...
    setTimeout(..., timerCallback, ...);
}

void timerCallback(...) {
    ...
}

int main() {
    send(.../* first message */);
    ... /* other initial setup */
    waitForAllTimeoutsAndMessagesThenExit(); // runs mainLoop
}
```

callback-based programming

writing scripts in a webpage

many graphical user interface libraries

sometimes servers that handle lots of connections

backup slides

BROKEN: producer/consumer signal

exercise: example why signal here is BROKEN? hint: two
consume()+two produce()

```
pthread_mutex_t lock; pthread_cond_t data_ready; UnboundedQueue buffer;
Produce(item) {
    pthread_mutex_lock(&lock);
    buffer.enqueue(item);
    /* GOOD CODE: pthread_cond_signal(&data_ready); */
    /* BAD CODE: */ if (buffer.size() == 1) pthread_cond_signal(&item);
    pthread_mutex_unlock(&lock);
}
Consume() {
    pthread_mutex_lock(&lock);
    while (buffer.empty()) {
        pthread_cond_wait(&data_ready, &lock);
    }
    item = buffer.dequeue();
    pthread_mutex_unlock(&lock);
    return item;
}
```

bad case (setup)

thread 0	1	2	3
Consume(): lock empty? wait on cv	Consume(): lock empty? wait on cv	Produce(): lock	Produce():

bad case

thread 0	1	2	3
Consume(): lock empty? wait on cv	Consume(): lock empty? wait on cv	Produce(): lock enqueue size = 1? signal unlock	Produce(): wait for lock gets lock enqueue size $\neq$ 1: don't signal unlock
wait for lock			
gets lock dequeue	still waiting		

monitor exercise: ConsumeTwo

suppose we want producer/consumer, but...

but change Consume() to ConsumeTwo() which returns a **pair of values**

and don't want two calls to ConsumeTwo() to wait...
with each getting one item

what should we change below?

```
pthread_mutex_t lock;  
pthread_cond_t data_ready;  
UnboundedQueue buffer;
```

```
Produce(item) {  
    pthread_mutex_lock(&lock);  
    buffer.enqueue(item);  
    pthread_cond_signal(&data_ready);  
    pthread_mutex_unlock(&lock);  
}
```

```
Consume() {  
    pthread_mutex_lock(&lock);  
    while (buffer.empty()) {  
        pthread_cond_wait(&data_ready, &lock);  
    }  
    item = buffer.dequeue();  
    pthread_mutex_unlock(&lock);  
    return item;  
}
```

monitor exercise: ordering

suppose we want producer/consumer, but...

but want to ensure first call to Consume() **always** returns first

(no matter what ordering cond_signal/cond_broadcast use)

```
pthread_mutex_t lock;  
pthread_cond_t data_ready;  
UnboundedQueue buffer;
```

```
Produce(item) {  
    pthread_mutex_lock(&lock);  
    buffer.enqueue(item);  
    pthread_cond_signal(&data_ready);  
    pthread_mutex_unlock(&lock);  
}
```

```
Consume() {  
    pthread_mutex_lock(&lock);  
    while (buffer.empty()) {  
        pthread_cond_wait(&data_ready, &lock);  
    }  
    item = buffer.dequeue();  
    pthread_mutex_unlock(&lock);  
    return item;  
}
```

the link layer

Ethernet, Wi-Fi, Bluetooth, DOCSIS (cable modems), ...

allows send/recv messages to machines on “same” network segment

- typically: wireless range+channel or connected to a single switch/router
- could be larger (if *bridging* multiple network segments)
- could be smaller (switch/router uses “virtual LANs”)

typically: source+destination specified with MAC addresses

- MAC = media access control

- usually manufacturer assigned / hard-coded into device
- unique address per port/wifi transmitter/etc.

can specify destination of “anyone” (called *broadcast*)

messages usually called “frames”

the link layer

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- unique address per port/wifi transmitter/etc.

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link layer jobs

divide raw bits into messages

identify who message is for on shared radio/wire

handle if two+ machines use radio/wire at same time

drop/resend messages if corruption detected

resending more common in radio schemes (wifi, etc.)

link layer reliability?

Ethernet + Wifi have checksums

Q1: Why doesn't this give us uncorrupted messages?

Why do we still have checksums at the higher layers?

Q2: What's a benefit of doing this if we're also doing it in the higher layer?

link layer quality of service

if frame gets...

event	on Ethernet	on WiFi
collides with another	detected + may resend	resend
not received	lose silently	resent
header corrupted	usually discard silently	usually resend
data corrupted	usually discard silently	usually resend
too long	not allowed to send	not allowed to send
reordered (v. other messages)	received out of order	received out of order
destination unknown	lose silently	usually resend??
too much being sent	discard excess?	discard excess?

network layer quality of service

if packet ...

event

on IPv4/v6

collides with another

out of scope — handled by link layer

not received

lost silently

header corrupted

usually discarded silently

data corrupted

received corrupted

too long

dropped with notice or “fragmented” + recombined

reordered (v. other messages)

received out of order

destination unknown

usually dropped with notice

too much being sent

discard excess

network layer quality of service

if packet ...

event

on IPv4/v6

collides with another

out of scope — handled by link layer

not received

lost silently

header corrupted

usually discarded silently

data corrupted

received corrupted

too long

dropped with notice or “fragmented” + recombined

reordered (v. other messages)

received out of order

destination unknown

usually dropped with notice

too much being sent

discard excess

includes dropped by link layer
(e.g. if detected corrupted there)

firewalls

don't want to expose network service to everyone?

solutions:

- service picky about who it accepts connections from
- filters in OS on machine with services
- filters on router

later two called “firewalls”

firewall rules examples?

ALLOW tcp port 443 (https) FROM everyone

ALLOW tcp port 22 (ssh) FROM my desktop's IP address

BLOCK tcp port 22 (ssh) FROM everyone else

ALLOW from address X to address Y

...

TCP state machine

TIME_WAIT, ESTABLISHED, ...?

OS tracks “state” of TCP connection

- am I just starting the connection?

- is other end ready to get data?

- am I trying to close the connection?

- do I need to resend something?

standardized set of state names

TIME_WAIT

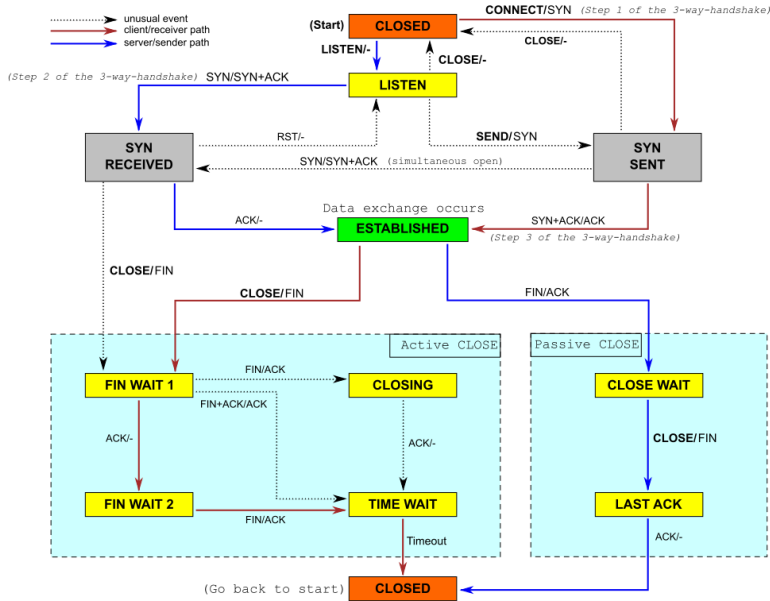
remember delayed messages?

problem for TCP ports

if I reuse port number, I can get message from old connection

solution: TIME_WAIT to make sure connection really done
done after sending last message in connection

TCP state machine picture



querying the root

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
edu.          172800      IN          NS          b.edu-servers.net.
edu.          172800      IN          NS          f.edu-servers.net.
edu.          172800      IN          NS          i.edu-servers.net.
edu.          172800      IN          NS          a.edu-servers.net.
...
b.edu-servers.net. 172800      IN          A           191.33.14.30
b.edu-servers.net. 172800      IN          AAAA        2001:503:231d::2:30
f.edu-servers.net. 172800      IN          A           192.35.51.30
f.edu-servers.net. 172800      IN          AAAA        2001:503:d414::30
...
;; Received 843 bytes from 198.97.190.53#53(h.root-servers.net) in 8 ms
...
```

querying the edu

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
```

virginia.edu.	172800	IN	NS	nom.virginia.edu.
virginia.edu.	172800	IN	NS	uvaarpa.virginia.edu.
virginia.edu.	172800	IN	NS	eip-01-aws.net.virginia.edu.
nom.virginia.edu.	172800	IN	A	128.143.107.101
uvaarpa.virginia.edu.	172800	IN	A	128.143.107.117
eip-01-aws.net.virginia.edu.	172800	IN	A	44.234.207.10

```
;; Received 165 bytes from 192.26.92.30#53(c.edu-servers.net) in 40 ms
```

```
...
```


querying virginia.edu+cs.virginia.edu

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
```

```
cs.virginia.edu.          3600      IN      NS      coresrv01.cs.virginia.edu.
```

```
coresrv01.cs.virginia.edu. 3600      IN      A      128.143.67.11
```

```
;; Received 116 bytes from 44.234.207.10#53(eip-01-aws.net.virginia.edu) in 72 ms
```

```
www.cs.Virginia.EDU.      172800    IN      A      128.143.67.11
```

```
cs.Virginia.EDU.          172800    IN      NS      coresrv01.cs.Virginia.EDU.
```

```
coresrv01.cs.Virginia.EDU. 172800    IN      A      128.143.67.11
```

```
;; Received 151 bytes from 128.143.67.11#53(coresrv01.cs.virginia.edu) in 4 ms
```

querying typical ISP's resolver

```
$ dig www.cs.virginia.edu
```

```
...
```

```
;; ANSWER SECTION:
```

```
www.cs.Virginia.EDU.          7183           IN           A           128.143.67.11
```

```
..
```

cached response

valid for 7183 more seconds

after that everyone needs to check again

'connected' UDP sockets

```
int fd = socket(AF_INET, SOCK_DGRAM, 0);
struct sockaddr_in my_addr= ...;
/* set local IP address + port */
bind(fd, &my_addr, sizeof(my_addr))
struct sockaddr_in to_addr = ...;
connect(fd, &to_addr); /* set remote IP address + port */
/* doesn't actually communicate with remote address yet */

...
int count = write(fd, data, data_size);
// OR
int count = send(fd, data, data_size, 0 /* flags */);
/* single message -- sent ALL AT ONCE */

int count = read(fd, buffer, buffer_size);
// OR
int count = recv(fd, buffer, buffer_size, 0 /* flags */);
/* receives whole single message ALL AT ONCE */
```

UDP sockets on IPv4

```
int fd = socket(AF_INET, SOCK_DGRAM, 0);
struct sockaddr_in my_addr= ...;
/* set local IP address + port */
if (0 != bind(fd, &my_addr, sizeof(my_addr)))
    handle_error();
...
struct sockaddr_in to_addr = ...;
/* send a message to specific address */
int bytes_sent = sendto(fd, data, data_size, 0 /* flags */,
    &to_addr, sizeof(to_addr));

struct sockaddr_in from_addr = ...;
/* receive a message + learn where it came from */
int bytes_recvd = recvfrom(fd, &buffer[0], buffer_size, 0,
    &from_addr, sizeof(from_addr));
...
```

what about non-local machines?

when configuring network specify:

range of addresses to expect on local network

128.148.67.0-128.148.67.255 on my desktop

“netmask”

gateway machine to send to for things outside my local network

128.143.67.1 on my desktop

my desktop looks up the corresponding MAC address

routes on my desktop

```
$ /sbin/route -n
```

```
Kernel IP routing table
```

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
0.0.0.0	128.143.67.1	0.0.0.0	UG	100	0	0	enp0s31f6
128.143.67.0	0.0.0.0	255.255.255.0	U	100	0	0	enp0s31f6
169.254.0.0	0.0.0.0	255.255.0.0	U	1000	0	0	enp0s31f6

network configuration says:

(line 2) to get to 128.143.67.0–128.143.67.255, send directly on local network

“genmask” is mask (for bitwise operations) to specify how big range is

(line 3) to get to 169.254.0.0–169.254.255.255, send directly on local network

(line 1) to get anywhere else, use “gateway” 128.143.67.1

names and addresses

name	address
logical identifier	location/how to locate
variable counter	memory address 0x7FFF9430
DNS name www.virginia.edu	IPv4 address 128.143.22.36
DNS name mail.google.com	IPv4 address 216.58.217.69
DNS name mail.google.com	IPv6 address 2607:f8b0:4004:80b::2005
DNS name reiss-t3620.cs.virginia.edu	IPv4 address 128.143.67.91
DNS name reiss-t3620.cs.virginia.edu	MAC address 18:66:da:2e:7f:da
service name https	port number 443
service name ssh	port number 22

two types of addresses?

MAC addresses: on link layer

IP addresses: on network layer

how do we know which MAC address to use?

a table on my desktop

my desktop:

```
$ arp -an
```

```
? (128.143.67.140) at 3c:e1:a1:18:bd:5f [ether] on enp0s31f6
```

```
? (128.143.67.236) at <incomplete> on enp0s31f6
```

```
? (128.143.67.11) at 30:e1:71:5f:39:10 [ether] on enp0s31f6
```

```
? (128.143.67.92) at <incomplete> on enp0s31f6
```

```
? (128.143.67.5) at d4:be:d9:b0:99:d1 [ether] on enp0s31f6
```

...

network address to link-layer address + interface

only tracks things directly connected to my local network

non-local traffic sent to local router

how is that table made?

ask all machines on local network (same switch)

“Who has 128.148.67.140”

the correct one replies

URLs and HTTP (1)

`http://www.foo.com:80/foo/bar?quux#q1`

lookup IP address of `www.foo.com`

connect via TCP to port 80:

`GET /foo/bar?quux HTTP/1.1`

`Host: www.foo.com:80`

URLs and HTTP (1)

`http://www.foo.com:80/foo/bar?quux#q1`

lookup IP address of `www.foo.com`

connect via TCP to port 80:

`GET /foo/bar?quux HTTP/1.1`

`Host: www.foo.com:80`

URLs and HTTP (1)

`http://www.foo.com:80/foo/bar?quux#q1`

lookup IP address of `www.foo.com`

connect via TCP to port 80:

`GET /foo/bar?quux HTTP/1.1`

`Host: www.foo.com:80`

exercise: why include the Host there?

querying the root

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
edu.          172800      IN          NS          b.edu-servers.net.
edu.          172800      IN          NS          f.edu-servers.net.
edu.          172800      IN          NS          i.edu-servers.net.
edu.          172800      IN          NS          a.edu-servers.net.
...
b.edu-servers.net. 172800      IN          A           191.33.14.30
b.edu-servers.net. 172800      IN          AAAA        2001:503:231d::2:30
f.edu-servers.net. 172800      IN          A           192.35.51.30
f.edu-servers.net. 172800      IN          AAAA        2001:503:d414::30
...
;; Received 843 bytes from 198.97.190.53#53(h.root-servers.net) in 8 ms
...
```

querying the edu

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
```

virginia.edu.	172800	IN	NS	nom.virginia.edu.
virginia.edu.	172800	IN	NS	uvaarpa.virginia.edu.
virginia.edu.	172800	IN	NS	eip-01-aws.net.virginia.edu.
nom.virginia.edu.	172800	IN	A	128.143.107.101
uvaarpa.virginia.edu.	172800	IN	A	128.143.107.117
eip-01-aws.net.virginia.edu.	172800	IN	A	44.234.207.10

```
;; Received 165 bytes from 192.26.92.30#53(c.edu-servers.net) in 40 ms
```

```
...
```

querying virginia.edu+cs.virginia.edu

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
```

```
cs.virginia.edu.          3600      IN      NS      coresrv01.cs.virginia.edu.
```

```
coresrv01.cs.virginia.edu. 3600      IN      A      128.143.67.11
```

```
;; Received 116 bytes from 44.234.207.10#53(eip-01-aws.net.virginia.edu) in 72 ms
```

```
www.cs.Virginia.EDU.      172800    IN      A      128.143.67.11
```

```
cs.Virginia.EDU.         172800    IN      NS      coresrv01.cs.Virginia.EDU.
```

```
coresrv01.cs.Virginia.EDU. 172800    IN      A      128.143.67.11
```

```
;; Received 151 bytes from 128.143.67.11#53(coresrv01.cs.virginia.edu) in 4 ms
```


querying typical ISP's resolver

```
$ dig www.cs.virginia.edu
```

```
...
```

```
;; ANSWER SECTION:
```

```
www.cs.Virginia.EDU.          7183           IN           A           128.143.67.11
```

```
..
```

cached response

valid for 7183 more seconds

after that everyone needs to check again

spoofing

if I only allow connections from my desktop's IP addresses,
how would you attack this?

hint: how do we know what address messages come from?

connection setup: server, manual

```
int server_socket_fd = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = INADDR_ANY; /* "any address I can use" */
    /* or: addr.s_addr.in_addr = INADDR_LOOPBACK (127.0.0.1) */
    /* or: addr.s_addr.in_addr = htonl(...); */
addr.sin_port = htons(9999); /* port number 9999 */

if (bind(server_socket_fd, &addr, sizeof(addr)) < 0) {
    /* handle error */
}
listen(server_socket_fd, MAX_NUM_WAITING);
...
int socket_fd = accept(server_socket_fd, NULL);
```

connection setup: server, manual

```
int server_socket_fd = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = INADDR_ANY; /* "any address I can use" */
/* or: addr.sin_addr.in_addr = INADDR_LOOPBACK (127.0.0.1) */
/* or: addr.sin_addr.in_addr = htonl(...); */
addr.sin_port = htons(9999); /* port number 9999 */
```

```
if (bind(server_socket_fd, &addr, sizeof(addr)) < 0) {
    /* handle error */
}
```

```
}
```

```
listen(server_socket_fd, 10);
```

```
int sock = accept(server_socket_fd, &addr, &len);
```

INADDR_ANY: accept connections for any address I can!
alternative: specify specific address

connection setup: server, manual

```
int server_socket_fd = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = INADDR_ANY; /* "any address I can use" */
/* or: addr.s_addr.in_addr = INADDR_LOOPBACK (127.0.0.1) */
/* or: addr.s_addr.in_addr = htonl(...); */
addr.sin_port = htons(9999); /* port number 9999 */

if (bind(server_socket_fd, &addr, sizeof(addr)) < 0) {
    /* handle error */
}
listen(server_socket_fd, 10);
int
```

bind to 127.0.0.1? only accept connections from same machine
what we recommend for FTP server assignment

connection setup: server, manual

```
int server_socket_fd = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = INADDR_ANY; /* "any address I can use" */
    /* or: addr.s_addr.in_addr = INADDR_LOOPBACK (127.0.0.1) */
    /* or: addr.s_addr.in_addr = htonl(...); */
addr.sin_port = htons(9999); /* port number 9999 */

if (bind(server_socket_fd, &addr, sizeof(addr)) < 0) {
    /* handle error */
}
listen(server_socket_fd, 10); choose the number of unaccepted connections
...
int socket_fd = accept(server_socket_fd, NULL);
```

connection setup: client — manual addresses

```
int sock_fd;

server = /* code on later slide */;
sock_fd = socket(
    AF_INET, /* IPv4 */
    SOCK_STREAM, /* byte-oriented */
    IPPROTO_TCP
);
if (sock_fd < 0) { /* handle error */ }

struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = htonl(2156872459); /* 128.143.67.11 */
addr.sin_port = htons(80); /* port 80 */
if (connect(sock_fd, (struct sockaddr*) &addr, sizeof(addr)) {
    /* handle error */
}
DoClientStuff(sock_fd); /* read and write from sock_fd */
close(sock_fd);
```

connection setup: client — manual addresses

```
int sock_fd;

server = /* code on later slide */;
sock_fd = socket(
    AF_INET, /* IPv4 */
    SOCK_STREAM, /* byte-oriented */
    IPPROTO_TCP
);
if (sock_fd < 0) { /* handle error */ }

// specify IPv4 instead of IPv6 or local-only sockets
// specify TCP (byte-oriented) instead of UDP ('datagram' oriented)
struct sockaddr_in addr;
addr.sin_addr.s_addr = htonl(2156872459); /* 128.143.67.11 */
addr.sin_port = htons(80); /* port 80 */
if (connect(sock_fd, (struct sockaddr*) &addr, sizeof(addr)) {
    /* handle error */
}
DoClientStuff(sock_fd); /* read and write from sock_fd */
close(sock_fd);
```


connection setup: client — manual addresses

```
int sock_fd;

server = /* code */
sock_fd = socket(AF_INET, /* SOCK_STREAM, /* byte-oriented */
                 IPPROTO_TCP
);
if (sock_fd < 0) { /* handle error */ }

struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = htonl(2156872459); /* 128.143.67.11 */
addr.sin_port = htons(80); /* port 80 */
if (connect(sock_fd, (struct sockaddr*) &addr, sizeof(addr)) {
    /* handle error */
}
DoClientStuff(sock_fd); /* read and write from sock_fd */
close(sock_fd);
```

htonl/s = host-to-network long/short
network byte order = big endian

connection setup: client — manual addresses

```
int sock_fd;
```

```
server = / struct representing IPv4 address + port number  
sock_fd = declared in <netinet/in.h>  
          AF_INET see man 7 ip on Linux for docs  
          SOCK_STREAM  
          IPPROTO_TCP
```

```
);  
if (sock_fd < 0) { /* handle error */ }
```

```
struct sockaddr_in addr;  
addr.sin_family = AF_INET;  
addr.sin_addr.s_addr = htonl(2156872459); /* 128.143.67.11 */  
addr.sin_port = htons(80); /* port 80 */  
if (connect(sock_fd, (struct sockaddr*) &addr, sizeof(addr)) {  
    /* handle error */  
}  
DoClientStuff(sock_fd); /* read and write from sock_fd */  
close(sock_fd);
```

echo client/server

```
void client_for_connection(int socket_fd) {
    int n; char send_buf[MAX_SIZE]; char recv_buf[MAX_SIZE];
    while (prompt_for_input(send_buf, MAX_SIZE)) {
        n = write(socket_fd, send_buf, strlen(send_buf));
        if (n != strlen(send_buf)) {...error?...}
        n = read(socket_fd, recv_buf, MAX_SIZE);
        if (n <= 0) return; // error or EOF
        write(STDOUT_FILENO, recv_buf, n);
    }
}



---


void server_for_connection(int socket_fd) {
    int read_count, write_count; char request_buf[MAX_SIZE];
    while (1) {
        read_count = read(socket_fd, request_buf, MAX_SIZE);
        if (read_count <= 0) return; // error or EOF
        write_count = write(socket_fd, request_buf, read_count);
        if (read_count != write_count) {...error?...}
    }
}
```

echo client/server

```
void client_for_connection(int socket_fd) {
    int n; char send_buf[MAX_SIZE]; char recv_buf[MAX_SIZE];
    while (prompt_for_input(send_buf, MAX_SIZE)) {
        n = write(socket_fd, send_buf, strlen(send_buf));
        if (n != strlen(send_buf)) {...error?...}
        n = read(socket_fd, recv_buf, MAX_SIZE);
        if (n <= 0) return; // error or EOF
        write(STDOUT_FILENO, recv_buf, n);
    }
}



---


void server_for_connection(int socket_fd) {
    int read_count, write_count; char request_buf[MAX_SIZE];
    while (1) {
        read_count = read(socket_fd, request_buf, MAX_SIZE);
        if (read_count <= 0) return; // error or EOF
        write_count = write(socket_fd, request_buf, read_count);
        if (read_count != write_count) {...error?...}
    }
}
```

echo client/server

```
void client_for_connection(int socket_fd) {
    int n; char send_buf[MAX_SIZE]; char recv_buf[MAX_SIZE];
    while (prompt_for_input(send_buf, MAX_SIZE)) {
        n = write(socket_fd, send_buf, strlen(send_buf));
        if (n != strlen(send_buf)) {...error?...}
        n = read(socket_fd, recv_buf, MAX_SIZE);
        if (n <= 0) return; // error or EOF
        write(STDOUT_FILENO, recv_buf, n);
    }
}

void server_for_connection(int socket_fd) {
    int read_count, write_count; char request_buf[MAX_SIZE];
    while (1) {
        read_count = read(socket_fd, request_buf, MAX_SIZE);
        if (read_count <= 0) return; // error or EOF
        write_count = write(socket_fd, request_buf, read_count);
        if (read_count != write_count) {...error?...}
    }
}
```

connection setup: server, address setup

```
/* example (hostname, portname) = ("127.0.0.1", "443") */
const char *hostname; const char *portname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;

memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_INET; /* for IPv4 */
/* or: */ hints.ai_family = AF_INET6; /* for IPv6 */
/* or: */ hints.ai_family = AF_UNSPEC; /* I don't care */
hints.ai_flags = AI_PASSIVE;

rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }
```

connection setup: server, address setup

```
/* example (hostname, portname) = ("127.0.0.1", "443") */
const char *hostname; const char *portname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;

memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_INET; /* for IPv4 */
/* or: */ hints.ai_family = AF_INET6; /* for IPv6 */
/* or: */ hints.ai_family = AF_UNSPEC; /* I don't care */
hints.ai_flags = AI_PASSIVE; /* hostname could also be NULL
                               means "use all possible addresses"
                               only makes sense for servers */
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) {
```

connection setup: server, address setup

```
/* example (hostname, portname) = ("127.0.0.1", "443") */
const char *hostname; const char *portname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;

memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_INET; /* for IPv4 */
/* or: */ hints.ai_family = AF_INET6; /* for IPv6 */
/* or: */ hints.ai_family = AF_UNSPEC; /* I don't care */
hints.ai_flags = 0;

rv = getaddrinfo(hostname, portname, &hints, NULL);
if (rv != 0) {
```

portname could also be NULL
means "choose a port number for me"
only makes sense for servers

connection setup: server, address setup

```
/* example (hostname, portname) = ("127.0.0.1", "443") */
const char *hostname = "127.0.0.1";
...
struct addrinfo *server;
struct addrinfo hints;
int rv;

memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_INET; /* for IPv4 */
/* or: */ hints.ai_family = AF_INET6; /* for IPv6 */
/* or: */ hints.ai_family = AF_UNSPEC; /* I don't care */
hints.ai_flags = AI_PASSIVE;

rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }
```

connection setup: server, addrinfo

```
struct addrinfo *server;
... getaddrinfo(...) ...

int server_socket_fd = socket(
    server->ai_family,
    server->ai_socktype,
    server->ai_protocol
);

if (bind(server_socket_fd, ai->ai_addr, ai->ai_addr_len)) < 0) {
    /* handle error */
}
listen(server_socket_fd, MAX_NUM_WAITING);
...
int socket_fd = accept(server_socket_fd, NULL);
```

connection setup: client, using addrinfo

```
int sock_fd;
struct addrinfo *server = /* code on next slide */;

sock_fd = socket(
    server->ai_family,
    // ai_family = AF_INET (IPv4) or AF_INET6 (IPv6) or ...
    server->ai_socktype,
    // ai_socktype = SOCK_STREAM (bytes) or ...
    server->ai_protocol,
    // ai_protocol = IPPROTO_TCP or ...
);
if (sock_fd < 0) { /* handle error */ }
if (connect(sock_fd, server->ai_addr, server->ai_addrlen) < 0) {
    /* handle error */
}
freeaddrinfo(server);
DoClientStuff(sock_fd); /* read and write from sock_fd */
close(sock_fd);
```

connection setup: client, using addrinfo

```
int sock_fd;  
struct addrinfo *server = /* code on next slide */;  
  
sock_fd = socket(  
    server->ai_family,  
    // ai_family = AF_INET (IPv4) or AF_INET6 (IPv6) or ...  
    server->ai_socktype,  
    // ai_socktype = SOCK_STREAM (bytes) or ...  
    server->ai_protocol,  
    // addrinfo contains all information needed to setup socket  
    // set by getaddrinfo function (next slide)  
);  
if (sock_fd < 0) {  
    if (errno == EAI_ADDRFAMILY) {  
        /* handles IPv4 and IPv6 */  
    }  
    /* handles DNS names, service names */  
}  
freeaddrinfo(server);  
DoClientStuff(sock_fd); /* read and write from sock_fd */  
close(sock_fd);
```

connection setup: client, using addrinfo

```
int sock_fd;
struct addrinfo *server = /* code on next slide */;

sock_fd = socket(
    server->ai_family,
    // ai_family = AF_INET (IPv4) or AF_INET6 (IPv6) or ...
    server->ai_socktype,
    // ai_socktype = SOCK_STREAM (bytes) or ...
    server->ai_protocol,
    // ai_protocol = IPPROTO_TCP or ...
);
if (sock_fd < 0) { /* handle error */ }
if (connect(sock_fd, server->ai_addr, server->ai_addrlen) < 0) {
    /* handle error */
}
freeaddrinfo(server);
DoClientStuff(sock_fd); /* read and write from sock_fd */
close(sock_fd);
```

connection setup: client, using addrinfo

```
int sock_fd;  
struct addrinfo *ai; /* ... */  
sock_fd = socket(server->ai_family, server->ai_socktype,  
    // ai_family = AF_INET (IPv4) or AF_INET6 (IPv6) or ...  
    server->ai_socktype,  
    // ai_socktype = SOCK_STREAM (bytes) or ...  
    server->ai_protocol  
    // ai_protocol = IPPROTO_TCP or ...  
);  
if (sock_fd < 0) { /* handle error */ }  
if (connect(sock_fd, server->ai_addr, server->ai_addrlen) < 0) {  
    /* handle error */  
}  
freeaddrinfo(server);  
DoClientStuff(sock_fd); /* read and write from sock_fd */  
close(sock_fd);
```

ai_addr points to struct representing address
type of struct depends whether IPv6 or IPv4

connection setup: client, using addrinfo

```
int sock_fd;
```

```
st
```

```
so
```

since addrinfo contains pointers to dynamically allocated memory,
call this function to free everything

```
    // ai_family = AF_INET (IPv4) or AF_INET6 (IPv6) or ...
    server->ai_socktype,
    // ai_socktype = SOCK_STREAM (bytes) or ...
    server->ai_protocol
    // ai_protocol = IPPROTO_TCP or ...
);
if (sock_fd < 0) { /* handle error */ }
if (connect(sock_fd, server->ai_addr, server->ai_addrlen) < 0) {
    /* handle error */
}
freeaddrinfo(server);
DoClientStuff(sock_fd); /* read and write from sock_fd */
close(sock_fd);
```

connection setup: lookup address

```
/* example hostname, portname = "www.cs.virginia.edu", "443" */
const char *hostname; const char *portname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;
memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_UNSPEC; /* for IPv4 OR IPv6 */
// hints.ai_family = AF_INET4; /* for IPv4 only */

hints.ai_socktype = SOCK_STREAM; /* byte-oriented --- TCP */
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }

/* eventually freeaddrinfo(result) */
```


connection setup: lookup address

```
/* example hostname, portname = "www.cs.virginia.edu", "443" */
const char *hostname; const char *portname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;
memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_UNSPEC; /* for IPv4 OR IPv6 */
// hints.ai_socktype = AF_INET; /* for TCP or UDP */
NB: pass pointer to pointer to addrinfo to fill in
hints.ai_socktype = SOCK_STREAM; /* byte-oriented --- TCP */
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }

/* eventually freeaddrinfo(result) */
```

connection setup: lookup address

```
/* example hostname, portname = "www.cs.virginia.edu", "443" */
const
...
struct
struct
int rv;
memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_UNSPEC; /* for IPv4 OR IPv6 */
// hints.ai_family = AF_INET4; /* for IPv4 only */

hints.ai_socktype = SOCK_STREAM; /* byte-oriented --- TCP */
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }

/* eventually freeaddrinfo(result) */
```

AF_UNSPEC: choose between IPv4 and IPv6 for me
AF_INET, AF_INET6: choose IPv4 or IPV6 respectively

connection setup: multiple server addresses

```
struct addrinfo *server;
...
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }

for (struct addrinfo *current = server; current != NULL;
     current = current->ai_next) {
    sock_fd = socket(current->ai_family, current->ai_socktype, current->ai_protocol);
    if (sock_fd < 0) continue;
    if (connect(sock_fd, current->ai_addr, current->ai_addrlen) == 0)
        break;
}
close(sock_fd); // connect failed
}
freeaddrinfo(server);
DoClientStuff(sock_fd);
close(sock_fd);
```

connection setup: multiple server addresses

```
struct addrinfo *server;
...
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }

for (struct addrinfo *current = server; current != NULL;
     current = current->ai_next) {
    sock_fd = socket(current->ai_family, current->ai_socktype, current->ai_protocol);
    if (sock_fd < 0) continue;
    if (connect(sock_fd, current->ai_addr, current->ai_addrlen) == 0)
        break;
}
close(sock_fd);
}
freeaddrinfo(server);
DoClientStuff(sock_fd);
close(sock_fd);
```

addrinfo is a linked list

name can correspond to multiple addresses

example: redundant copies of web server

example: an IPv4 address and IPv6 address

example: wired + wireless connection on one machine

connection setup: old lookup function

```
/* example hostname, portnum= "www.cs.virginia.edu", 443*/
const char *hostname; int portnum;
...
struct hostent *server_ip;
server_ip = gethostbyname(hostname);

if (server_ip == NULL) { /* handle error */ }

struct sockaddr_in addr;
addr.s_addr = *(struct in_addr*) server_ip->h_addr_list[0];
addr.sin_port = htons(portnum);
sock_fd = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
connect(sock_fd, &addr, sizeof(addr));
...
```

aside: on server port numbers

Unix convention: must be root to use ports 0–1023

root = superuser = 'administrator user' = what sudo does

so, for testing: probably ports > 1023